Mathematics

Vrije Universiteit Amsterdam - Faculty of Science - Mathematics - 2025-2026

Mathematics is a vibrant and versatile field. In research and teaching, the Vrije Universiteit Amsterdam focuses on both pure and applied mathematics. We perform research in algebra, number theory, geometry, analysis, dynamical systems, probability, statistics, and operations research.

Abstract mathematical theory often finds its way to unexpected applications. For example, modern cryptography makes essential use of prime numbers, bacterial growth is governed by differential equations, and the spread of forest fires and infections can be predicted by stochastic models. At VU Amsterdam we apply mathematical theory in business analytics, big data, forensic statistics, systems biology, neuroscience, and complex systems.

During the first year, you will participate in the “Introduction to Master Mathematics". In this course, you will learn about the teaching and research activities of the various groups in the Mathematics Department, possible thesis projects, and job prospects. Throughout your studies, you will specialise in an area of particular interest by choosing one of the study paths. The last semester is dedicated to the Master’s Project, which is either a research project under the supervision of a staff member, or an external internship at a company or research facility. In the latter case, you will have an on-location advisor and a supervisor from the Department of Mathematics.

Within the specialisation Applied Mathematics, the programme offers two study paths: Biomedical Mathematics, and Data Science and Optimization. Within the specialisation Pure Mathematics, the programme offers three study paths: Algebra, Geometry and Number Theory, Analysis and Dynamical Systems, and Probability and Statistics. You can also choose the Education specialisation, where the teaching qualification part is taught in Dutch and you obtain the qualification to teach all levels of high school ('eerstegraads lesbevoegdheid wiskunde').

Info

Level	Master
Language	English
Duration	2 years
Type	Full and Part Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Mathematics specialisation Applied Mathematics

Description

Within the specialisation Applied Mathematics you can choose between two study paths: Biomedical Mathematics or Data Science and Optimization.

Your programme has a workload of 120 EC and is structured as follows:

- there are 54 EC of compulsory components common to both study paths in Applied Mathematics (see the tab specialisation Applied Mathematics compulsory courses),
- there are seminars, common to all study paths in Applied and Pure Mathematics, of which you must follow 2 or 3 (see the tab Mathematics seminars),
- there is a compulsory course that is specific to your study path (see the tab for your study path),
- there is a constrained choice that is specific to your study path (see the tab for your study path),
- the remainder of your programme (up to 16 EC for Biomedical Mathematics and up to 36 EC for Data Science and Optimization) you fill with electives (see the tab Mathematics electives).

As for electives, further note that:

- Within the tab for your study path, you will find a tab with *suggested* electives. Compared to the tab Mathematics electives, the suggested electives are a subset of *natural* choices for your study path. You do not have to restrict your choice to the suggested electives - it is a non-binding recommendation.
- Up to *12 EC* of your electives can be *any* courses *of level 400 or higher* that are not linked/listed under the tab Mathematics electives. These can be courses outside of mathematics, e.g. in literature. A prior consent from the Examination Board is required.

Programme Sections

- Mathematics specialisation Applied Mathematics compulsory courses
- Mathematics Biomedical Mathematics study path
- Mathematics seminars
- Mathematics Data Science and Optimization study path
- Mathematics electives (MasterMath and Local courses)

Mathematics specialisation Applied Mathematics compulsory courses

Description

The following components are compulsory in both study paths of the specialisation Applied Mathematics.

Courses

Course Name	Period	Credits	Code
Introduction to Master Mathematics	Ac. Year (September)	6EC	XM_0095
Master Project Mathematics	Ac. Year (September)	36EC	X_400355
Advanced Machine Learning	P1	6EC	XM_0010
Statistical Models	P1+2	6EC	X_400418

Mathematics Biomedical Mathematics study path

Programme Sections

- Mathematics Biomedical Mathematics compulsory courses
- Mathematics Biomedical Mathematics constrained choice
- Mathematics Biomedical Mathematics suggested electives

Mathematics Biomedical Mathematics compulsory courses

Description

In the study path Biomedical Mathematics, the course Mathematical Biology is compulsory.

* Mathematical Biology is taught every other year. It is given in 2025-26 and will not be given in 2026-27.

Courses

Course Name	Period	Credits	Code
Mathematical Biology	P1+2+3	8EC	XMM_400504

Mathematics Biomedical Mathematics constrained choice

Description

Constrained choice for the study path Biomedical Mathematics: choose at least 30 EC from the list of life science courses below.

1. Into your 30 EC package of life science courses, you can include courses of level 400 or higher that are outside of the list below. A prior consent from the Examination Board is required.

2. The courses From Protein to Cell, Metabolomics are of level 300. At most two level-300 courses are allowed in your programme.

3. Note that the courses Brain Imaging for AI and Introduction to Systems Biology are not intended for students who have previously followed, respectively, P_MBRIMAG Brain Imaging and X_428566 Principles of Systems Biology, due to a considerable overlap.

Courses

Course Name	Period	Credits	Code
From Molecule to Mind	P1	6EC	AM_1275
Fundamentals of Bioinformatics	P1	6EC	X_405052
Introduction to Systems Biology	P1	6EC	X_428565
Algorithms in Sequence Analysis	P2	6EC	X_405050
From Protein to Cell	P2	6EC	AB_1052
Mechanics and Thermodynamics in the Cell	P2	6EC	X_422589
Metabolomics	P2	6EC	AB_1291
Basic Models of Biological Networks	P4	6EC	X_418154
Brain Imaging	P4	6EC	P_MBRIMAG_AI
Structural Bioinformatics	P4	6EC	X_405019
Bioinformatics for Translational Medicine	P5	6EC	X_405092
Tracer Kinetic Modelling	P5	6EC	XM_0014

Mathematics Biomedical Mathematics suggested electives

Description

The courses listed below are natural choices of electives for the study path Biomedical Mathematics. The majority of the courses on this list, except for Functional Analysis and Inverse Problems in Imaging, are well connected to the research at the VU Mathematics department and will prepare you particularly well for your Master Project. This is a non-binding recommendation.

Courses

Course Name	Period	Credits	Code
Dynamical Systems	P1+2+3	8EC	XMM_400429
Functional Analysis	P1+2+3	8EC	XMM_400328
Measure Theoretical Probability	P1+2+3	8EC	XMM_400244
Numerical Methods for PDEs	P1+2+3	8EC	XMM_0072
Partial Differential Equations	P1+2+3	8EC	XMM_400330
Bifurcation Theory	P4+5+6	8EC	XMM_0078
Inverse Problems in Imaging	P4+5+6	6EC	XMM_0047
Machine Learning Theory	P4+5+6	8EC	XMM_0002
Stochastic Processes	P4+5+6	8EC	XMM_400339

Mathematics seminars

Description

From 2025-26 on, the programme will offer 8 seminars on an alternate basis. Those students who have started in 2025-26, must complete 2 or 3 out of these 8 seminars.*

In 2025-26 and every other year after that, the 4 seminars linked under this tab will be taught.

For 2026-27 and every other year after that, the following 4 seminars are tentatively planned:

- Analysis of Neuroimaging Data,
- Mathematical Optimization,
- Algebraic Topology and Dynamical Systems,
- Concentration Inequalities.

As a non-binding recommendation, the following seminar choices are especially suitable for the study path

- Biomedical Mathematics: Network Dynamics, Symmetry and Synchrony, Sequential Monte Carlo Methods, Uncertainty Quantification, Analysis of Neuroimaging Data;
- Data Science and Optimization: Sequential Monte Carlo Methods, Uncertainty Quantification, Mathematical Optimization, Concentration Inequalities;
- Analysis and Dynamical Systems: Network Dynamics, Symmetry and Synchrony, Uncertainty Quantification, Algebraic Topology and Dynamical Systems, Concentration Inequalities;
- Algebra, Geometry, and Number Theory: p-adic Numbers and Their Applications, Algebraic Topology and Dynamical Systems;
- Probability and Statistics: Sequential Monte Carlo Methods, Uncertainty Quantification, Analysis of Neuroimaging Data, Concentration Inequalities.

* Those students who have started before or in 2024-25 and have completed 1 or 2 of the discontinued

- XM_41013 Master Seminar in Analysis and Dynamical Systems,
- XM_0058 Master Seminar in Algebra and Geometry,
- XM_41011 Master Seminar in Stochastics,

are exempt from the requirement to follow any of the 8 new seminars. However, if they wish to, they are allowed to choose a 2nd and/or 3rd seminar among the 8 new ones except (the upcoming in 2026-27) Algebraic Topology and Dynamical Systems will not be allowed in combination with (the discontinued) XM_41013 Master Seminar in Analysis and Dynamical Systems, due to a considerable overlap.

Courses

Course Name	Period	Credits	Code
p-Adic Numbers and Their Applications	P1+2	6EC	XM_0184
Uncertainty Quantification	P1+2	6EC	XM_0183
Network Dynamics, Symmetry and Synchrony	P4+5	6EC	XM_0181
Sequential Monte Carlo Methods	P4+5	6EC	XM_0182

Mathematics Data Science and Optimization study path

Programme Sections

- Mathematics Data Science and Optimization compulsory courses
- Mathematics Data Science and Optimization constrained choice
- Mathematics Data Science and Optimization suggested electives

Mathematics Data Science and Optimization compulsory courses

Description

In the study path Data Science and Optimization, the course Mathematical Optimization iscompulsory.

Courses

Course Name	Period	Credits	Code
Mathematical Optimization	P2	6EC	XM_0051

Mathematics Data Science and Optimization constrained choice

Description

Constrained choice for the study path Data Science and Optimization: choose at least two courses from this list.

Courses

Course Name	Period	Credits	Code
Applied Stochastic Modeling	P1+2	6EC	X_400392
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Optimization of Business Processes	P4+5	6EC	X_400422

Mathematics Data Science and Optimization suggested electives

Description

The courses listed below are natural choices of electives for the study path Data Science and Optimization. The majority of the courses on this list, except for Discrete Optimization, Advanced Discrete Optimisation, Semidefinite Optimization, and Multivariate Statistics are well connected to the research at the VU Mathematics department and will prepare you particularly well for your Master Project. This is a non-binding recommendation.

Courses

Course Name	Period	Credits	Code
Asymptotic Statistics	P1+2+3	8EC	XMM_400323
Continuous Optimization	P1+2+3	6EC	XMM_400446
Discrete Optimization	P1+2+3	6EC	XMM_400445
Measure Theoretical Probability	P1+2+3	8EC	XMM_400244
Multivariate Statistics	P1+2+3	8EC	XMM_0081
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Performance of Networked Systems	P2	6EC	X_405105
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Advanced Discrete Optimisation	P4+5+6	6EC	XMM_0075
Bayesian Statistics	P4+5+6	8EC	XMM_40004
Machine Learning Theory	P4+5+6	8EC	XMM_0002
Queueing Theory	P4+5+6	6EC	XMM_400397
Scheduling	P4+5+6	6EC	XMM_400396
Semidefinite Optimization	P4+5+6	8EC	XMM_0015
Stochastic Integration	P4+5+6	8EC	XMM_0049

Mathematics electives (MasterMath and Local courses)

Description

Without prior consent from the Examination Board, the student can take any of the (level-400 or higher) mathematics electives linked under this tab.

* XMM_0072 Numerical Methods for PDEs is not intended for students who have previously followed XMM_0028 Numerical Methods for Time-dependent PDEs and/or XMM_0046 Numerical Methods for Stationary PDEs, due to a considerable overlap.

** XMM_0080 Interacting Particle Systems in the Lattice and Continuum is not intended for students who have previously followed XMM_0064 Statistical Mechanics, due to a considerable overlap.

*** XM_0140 Applied Analysis: Financial Mathematics (MSc) is not intended for students who have previously followed X_400076 Applied Analysis: Financial Mathematics, due to a considerable overlap.

Additionally, without prior consent from the Examination Board but subject to the constraint of *at most two level-300 courses in all study paths of the programme except for the Teachers study path*, the student can take any of the following level-300 mathematics electives (to view the studyguides of these courses, search by course code or name on studiegids.vu.nl):

- X_400631 Differential Geometry (BSc),
- XB_0068 Differential Topology,
- XB_0079 Functional Analysis,
- XB_0134 Galois Theory,
- XB_0135 Markov Chains,
- XB_0136 Mathematical Statistics 2,
- X_401028 Measure Theory,
- XB_0098 Non-linear Dynamical Systems****,
- X_400632 Number Theory,
- X_400163 Partial Differential Equations (BSc).

**** is not intended for students who have previously followed the discontinued XM_0169 Non-linear Dynamical Systems (MSc), due to a considerable overlap.

Finally, note that, in all study paths of the programme except for the Teachers study path, the student can choose *up to 12 EC* of electives outside of the courses linked/listed here. These can be courses outside of mathematics, e.g. in literature. In the study paths in Applied/Pure Mathematics, these must be courses of *level 400 or higher*. In the Education study path, these may also be the following courses from the Beta4all/Wisk4all programme: Meetkunde (6 EC), Geschiedenis (6 EC). Advance permission must be obtained in writing from the Examination Board.

Courses

Course Name	Period	Credits	Code
Advanced Machine Learning	P1	6EC	XM_0010
Applied Stochastic Modeling	P1+2	6EC	X_400392
Statistical Models	P1+2	6EC	X_400418
Stochastic Processes for Finance	P1+2	6EC	X_400352
Advanced Algebraic Geometry	P1+2+3	8EC	XMM_418083
Algebraic Geometry 1	P1+2+3	8EC	XMM_40001
Algebraic Number Theory	P1+2+3	8EC	XMM_400324
Algebraic Topology 1	P1+2+3	8EC	XMM_400600
Analytic Number Theory	P1+2+3	8EC	XMM_0153
Asymptotic Statistics	P1+2+3	8EC	XMM_400323
Category Theory	P1+2+3	8EC	XMM_418114
Commutative Algebra	P1+2+3	8EC	XMM_418136
Complexity Theory	P1+2+3	8EC	XMM_0006
Continuous Optimization	P1+2+3	6EC	XMM_400446
Differential Geometry	P1+2+3	8EC	XMM_400509
Discrete Optimization	P1+2+3	6EC	XMM_400445
Dynamical Systems	P1+2+3	8EC	XMM_400429
Forensic Probability and Statistics	P1+2+3	8EC	XMM_0005
Functional Analysis	P1+2+3	8EC	XMM_400328
High-dimensional Probability	P1+2+3	8EC	XMM_0079
Infinity-categories	P1+2+3	8EC	XMM_0062
Lie Groups	P1+2+3	8EC	XMM_400350
Mathematical Biology	P1+2+3	8EC	XMM_400504
Measure Theoretical Probability	P1+2+3	8EC	XMM_400244
Modern Cryptography	P1+2+3	8EC	XMM_0053
Modular Forms	P1+2+3	8EC	XMM_400599
Multivariate Statistics	P1+2+3	8EC	XMM_0081
Numerical Linear Algebra	P1+2+3	8EC	XMM_400329
Numerical Methods for PDEs	P1+2+3	8EC	XMM_0072
Partial Differential Equations	P1+2+3	8EC	XMM_400330
Quantum Computing	P1+2+3	8EC	XMM_0013
Quantum Information Theory	P1+2+3	8EC	XMM_0024
Systems and Control	P1+2+3	6EC	XMM_400332
Algorithms in Sequence Analysis	P2	6EC	X_405050
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Mathematical Optimization	P2	6EC	XM_0051
Performance of Networked Systems	P2	6EC	X_405105
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Optimization of Business Processes	P4+5	6EC	X_400422

Course Name	Period	Credits	Code
Additive Combinatorics	P4+5+6	8EC	XMM_0025
Advanced Discrete Optimisation	P4+5+6	6EC	XMM_0075
Advanced Numerical Linear Algebra with Applications	P4+5+6	8EC	XMM_0076
Advanced Representation Theory and Applications	P4+5+6	8EC	XMM_0077
Algebraic Geometry 2	P4+5+6	8EC	XMM_40002
Algebraic Topology 2	P4+5+6	8EC	XMM_0017
Bayesian Statistics	P4+5+6	8EC	XMM_40004
Bifurcation Theory	P4+5+6	8EC	XMM_0078
Causality	P4+5+6	8EC	XMM_0057
Coding Theory	P4+5+6	8EC	XMM_418160
Computational Finance	P4+5+6	8EC	XMM_0060
Elliptic Curves	P4+5+6	8EC	XMM_400505
Formal Methods in Mathematics	P4+5+6	8EC	XMM_0066
Graph Symmetries and Combinatorial Designs	P4+5+6	8EC	XMM_0044
Homotopy Type Theory	P4+5+6	8EC	XMM_0061
Interacting Particle Systems in the Lattice and Continuum	P4+5+6	8EC	XMM_0080
Inverse Problems in Imaging	P4+5+6	6EC	XMM_0047
Lie Algebras	P4+5+6	8EC	XMM_0037
Machine Learning Theory	P4+5+6	8EC	XMM_0002
Numerical Methods for Stochastic Differential Equations	P4+5+6	8EC	XMM_0071
Operator Algebras	P4+5+6	8EC	XMM_418062
Optimal Transport	P4+5+6	8EC	XMM_0063
Queueing Theory	P4+5+6	6EC	XMM_400397
Riemann Surfaces	P4+5+6	8EC	XMM_400325
Scheduling	P4+5+6	6EC	XMM_400396
Selected Areas in Cryptology	P4+5+6	8EC	XMM_40011
Semidefinite Optimization	P4+5+6	8EC	XMM_0015
Set Theory	P4+5+6	8EC	XMM_418035
Stochastic Integration	P4+5+6	8EC	XMM_0049
Stochastic Processes	P4+5+6	8EC	XMM_400339
Symplectic Geometry	P4+5+6	8EC	XMM_418036
Topological Data Analysis	P4+5+6	8EC	XMM_0036

Mathematics specialisation Mathematics Education

Description

* Dutch is required as the teaching qualification part of this programme is in Dutch.

There are two study paths within the specialisation Mathematics Education:

- the Education study path, for anybody admissible to the master's programme Mathematics who wishes to obtain an "eerstegraads lesbevoegdheid" and is not a HBO student with a "tweedegraads lesbevoegdheid",
- the Teachers study path, specifically for HBO students from the "lerarenopleiding wiskunde" with a "tweedegraads lesbevoegdheid", who can enter the study path after having followed a premaster programme.

Either study path consists of mathematical courses and a training towards an "eerstegraads lesbevoegdheid wiskunde". The teaching qualification training includes didactical components, work experience through an internship at a secondary school, and a practice-based research project. If you follow the Education study path, mathematical courses and the teaching qualification training form equal parts of your programme. If you follow the Teachers study path, the teaching qualification training is reduced and the larger part of your programme is aimed at obtaining knowledge of mathematics, eventually at the master level. In either study path, your final master's thesis consists of a literature survey on an aspect of mathematical theory.

Programme Sections

- Secondary Education Teacher in Science - Specialization Mathematics
- Mathematics Education study path
- Mathematics electives (MasterMath and Local courses)
- Mathematics Teachers study path

Secondary Education Teacher in Science - Specialization Mathematics

Description

Ben jij wiskundig sterk én niet bang om dingen uit te proberen? En heb je oprechte belangstelling voor de manier waarop jongeren redeneren?

In een tijd van rekenmachines, computers en rekentools is het extra belangrijk dat een mens zelf inzicht heeft in formules, logisch kan redeneren en met een wiskundige bril naar de wereld kan kijken. Iemand die analytisch is, kijkt op een exacte manier naar de wereld om zich heen. Als leraar Wiskunde stimuleer je leerlingen om zelf na te denken en conclusies te trekken. Daar heeft iemand een leven langs iets aan.

Programme Sections

- Leraar VHO februari start
- Leraar VHO September start

Leraar VHO februari start

Courses

Course Name	Period	Credits	Code
Pedagogische en Didactische verdieping Semester 1	P1+2+3	9EC	O_MPEDIVER_F
Peergroup 2 Semester 1	P1+2+3	0EC	O_MLPEERG_2F
Practice 3	P1+2+3	15EC	O_MLPRAK_3F
Didactics 1	P4	6EC	O_MLDIDAC_1F
Practice 1	P4	6EC	O_MLPRAK_1F
Peergroup 1 Semester 2	P4+5+6	0EC	O_MLPEERG_1F
Didactiek 2 Semester 2	P5+6	9EC	O_MLDIDAC2F
Practice 2	P5+6	9EC	O_MLPRAK_2F

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Leraar VHO September start

Courses

Course Name	Period	Credits	Code
Didactics 1	P1	6EC	O_MLDIDAC_1
Practice 1	P1	6EC	O_MLPRAK_1
Peergroup 1 Semester 1	P1+2+3	0EC	O_MLPEERGR_1
Didactiek 2 Semester 1	P2+3	9EC	O_MLDIDAC2

Course Name	Period	Credits	Code
Practice 2	P2+3	9EC	O_MLPRAK_2
Pedagogische en Didactische Verdieping Semester 2	P4+5+6	9EC	O_MPEDIVER
Peergroup 2 Semester 2	P4+5+6	0EC	O_MLPEERGR_2
Practice 3	P4+5+6	15EC	O_MLPRAK_3

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Mathematics Education study path

Description

The programme of the Education study path has a 120 EC workload and includes:

- Master Leraar VHO Wiskunde (60 EC),
- Master Project Mathematics (T,E track), compulsory (24 EC),
- the rest (36 EC) is filled with electives (see the tab Mathematics electives).

Further note that

- If you have done the Educational minor in your bachelor, you will be exempted from some parts - 27 EC worth - of Master Leraar VHO Wiskunde.
- Up to *12 EC* of your electives can be *any* courses *of level 400 or higher* that are not linked/listed under the tab Mathematics electives. These can be courses outside of mathematics, e.g. in literature. These can also be the following courses from the Beta4all/Wisk4all programme: Meetkunde (6 EC), Geschiedenis (6 EC). A prior consent from the Examination Board is required.

Courses

Course Name	Period	Credits	Code
Master Project Mathematics (T,E track)	Ac. Year (September)	24EC	X_405037

Mathematics electives (MasterMath and Local courses)

Description

Without prior consent from the Examination Board, the student can take any of the (level-400 or higher) mathematics electives linked under this tab.

* XMM_0072 Numerical Methods for PDEs is not intended for students who have previously followed XMM_0028 Numerical Methods for Time-dependent PDEs and/or XMM_0046 Numerical Methods for Stationary PDEs, due to a considerable overlap.

** XMM_0080 Interacting Particle Systems in the Lattice and Continuum is not intended for students who have

previously followed XMM_0064 Statistical Mechanics, due to a considerable overlap.

*** XM_0140 Applied Analysis: Financial Mathematics (MSc) is not intended for students who have previously followed X_400076 Applied Analysis: Financial Mathematics, due to a considerable overlap.

Additionally, without prior consent from the Examination Board but subject to the constraint of *at most two level-300 courses in all study paths of the programme except for the Teachers study path*, the student can take any of the following level-300 mathematics electives (to view the studyguides of these courses, search by course code or name on studiegids.vu.nl):

- X_400631 Differential Geometry (BSc),
- XB_0068 Differential Topology,
- XB_0079 Functional Analysis,
- XB_0134 Galois Theory,
- XB_0135 Markov Chains,
- XB_0136 Mathematical Statistics 2,
- X_401028 Measure Theory,
- XB_0098 Non-linear Dynamical Systems****,
- X_400632 Number Theory,
- X_400163 Partial Differential Equations (BSc).

**** is not intended for students who have previously followed the discontinued XM_0169 Non-linear Dynamical Systems (MSc), due to a considerable overlap.

Finally, note that, in all study paths of the programme except for the Teachers study path, the student can choose *up to 12 EC* of electives outside of the courses linked/listed here. These can be courses outside of mathematics, e.g. in literature. In the study paths in Applied/Pure Mathematics, these must be courses of *level 400 or higher*. In the Education study path, these may also be the following courses from the Beta4all/Wisk4all programme: Meetkunde (6 EC), Geschiedenis (6 EC). Advance permission must be obtained in writing from the Examination Board.

Courses

Course Name	Period	Credits	Code
Advanced Machine Learning	P1	6EC	XM_0010
Applied Stochastic Modeling	P1+2	6EC	X_400392
Statistical Models	P1+2	6EC	X_400418
Stochastic Processes for Finance	P1+2	6EC	X_400352
Advanced Algebraic Geometry	P1+2+3	8EC	XMM_418083
Algebraic Geometry 1	P1+2+3	8EC	XMM_40001
Algebraic Number Theory	P1+2+3	8EC	XMM_400324
Algebraic Topology 1	P1+2+3	8EC	XMM_400600
Analytic Number Theory	P1+2+3	8EC	XMM_0153
Asymptotic Statistics	P1+2+3	8EC	XMM_400323
Category Theory	P1+2+3	8EC	XMM_418114
Commutative Algebra	P1+2+3	8EC	XMM_418136
Complexity Theory	P1+2+3	8EC	XMM_0006
Continuous Optimization	P1+2+3	6EC	XMM_400446
Differential Geometry	P1+2+3	8EC	XMM_400509
Discrete Optimization	P1+2+3	6EC	XMM_400445
Dynamical Systems	P1+2+3	8EC	XMM_400429
Forensic Probability and Statistics	P1+2+3	8EC	XMM_0005
Functional Analysis	P1+2+3	8EC	XMM_400328
High-dimensional Probability	P1+2+3	8EC	XMM_0079
Infinity-categories	P1+2+3	8EC	XMM_0062
Lie Groups	P1+2+3	8EC	XMM_400350
Mathematical Biology	P1+2+3	8EC	XMM_400504
Measure Theoretical Probability	P1+2+3	8EC	XMM_400244
Modern Cryptography	P1+2+3	8EC	XMM_0053
Modular Forms	P1+2+3	8EC	XMM_400599
Multivariate Statistics	P1+2+3	8EC	XMM_0081
Numerical Linear Algebra	P1+2+3	8EC	XMM_400329
Numerical Methods for PDEs	P1+2+3	8EC	XMM_0072

Course Name	Period	Credits	Code
Partial Differential Equations	P1+2+3	8EC	XMM_400330
Quantum Computing	P1+2+3	8EC	XMM_0013
Quantum Information Theory	P1+2+3	8EC	XMM_0024
Systems and Control	P1+2+3	6EC	XMM_400332
Algorithms in Sequence Analysis	P2	6EC	X_405050
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Mathematical Optimization	P2	6EC	XM_0051
Performance of Networked Systems	P2	6EC	X_405105
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Optimization of Business Processes	P4+5	6EC	X_400422
Additive Combinatorics	P4+5+6	8EC	XMM_0025
Advanced Discrete Optimisation	P4+5+6	6EC	XMM_0075
Advanced Numerical Linear Algebra with Applications	P4+5+6	8EC	XMM_0076
Advanced Representation Theory and Applications	P4+5+6	8EC	XMM_0077
Algebraic Geometry 2	P4+5+6	8EC	XMM_40002
Algebraic Topology 2	P4+5+6	8EC	XMM_0017
Bayesian Statistics	P4+5+6	8EC	XMM_40004
Bifurcation Theory	P4+5+6	8EC	XMM_0078
Causality	P4+5+6	8EC	XMM_0057
Coding Theory	P4+5+6	8EC	XMM_418160
Computational Finance	P4+5+6	8EC	XMM_0060
Elliptic Curves	P4+5+6	8EC	XMM_400505
Formal Methods in Mathematics	P4+5+6	8EC	XMM_0066
Graph Symmetries and Combinatorial Designs	P4+5+6	8EC	XMM_0044
Homotopy Type Theory	P4+5+6	8EC	XMM_0061
Interacting Particle Systems in the Lattice and Continuum	P4+5+6	8EC	XMM_0080
Inverse Problems in Imaging	P4+5+6	6EC	XMM_0047
Lie Algebras	P4+5+6	8EC	XMM_0037
Machine Learning Theory	P4+5+6	8EC	XMM_0002
Numerical Methods for Stochastic Differential Equations	P4+5+6	8EC	XMM_0071
Operator Algebras	P4+5+6	8EC	XMM_418062
Optimal Transport	P4+5+6	8EC	XMM_0063
Queueing Theory	P4+5+6	6EC	XMM_400397
Riemann Surfaces	P4+5+6	8EC	XMM_400325
Scheduling	P4+5+6	6EC	XMM_400396
Selected Areas in Cryptology	P4+5+6	8EC	XMM_40011
Semidefinite Optimization	P4+5+6	8EC	XMM_0015
Set Theory	P4+5+6	8EC	XMM_418035
Stochastic Integration	P4+5+6	8EC	XMM_0049
Stochastic Processes	P4+5+6	8EC	XMM_400339
Symplectic Geometry	P4+5+6	8EC	XMM_418036
Topological Data Analysis	P4+5+6	8EC	XMM_0036

Mathematics Teachers study path

Description

The programme of the Teachers study path has a 120 EC workload and includes:

- Master Leraar VHO Wiskunde (30 out of 60 EC are exempted),
- a selection of courses from the bachelor Mathematics programme, compulsory (42 EC, see below),
- Master Project Mathematics (T,E track), compulsory (24 EC),
- the rest (24 EC) is filled with the *level-400 and higher* electives from the tab Mathematics electives.

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Master Project Mathematics (T,E track)	Ac. Year (September)	24EC	X_405037
Fourier Analysis	P1+2	6EC	XB_0005
Mathematical Statistics	P1+2	6EC	XB_0049
Measure Theory	P1+2	6EC	X_401028
Numerical Methods	P1+2	6EC	X_401039
Complex Analysis	P4+5	6EC	X_400386
Dynamical Systems	P4+5	6EC	X_400637
Statistical Data Analysis	P4+5	6EC	X_401029

Mathematics specialisation Pure Mathematics

Description

Within the specialisation Pure Mathematics you can choose between three study paths: Algebra, Geometry and Number Theory, Analysis and Dynamical Systems, or Probability and Statistics.

Your programme has a workload of 120 EC and is structured as follows:

- Introduction to Master Mathematics and Master Project Mathematics are compulsory for all study paths in Pure and Applied Mathematics (see the tab specialisation Pure Mathematics compulsory courses),
- there are seminars, common to all study paths in Applied and Pure Mathematics, of which you must follow two or three (see the tab Mathematics seminars),
- there are compulsory / constrained choice courses that are specific to your study path (see the tab for your study path),
- the remainder of your programme (up to 42 EC) you fill with electives (see the tab Mathematics electives).

As for electives, further note that:

- Within the tab for your study path, you will find a tab with *suggested* electives. Compared to the tab Mathematics electives, the suggested electives are a subset of *natural* choices for your study path. You do not have to restrict your choice to the suggested electives - it is a non-binding recommendation.
- Up to *12 EC* of your electives can be *any* courses *of level 400 or higher* that are not linked/listed under the tab Mathematics electives. These can be courses outside of mathematics, e.g. in literature. A prior consent from the Examination Board is required.

Programme Sections

- Mathematics Analysis and Dynamical Systems study path
- Mathematics Algebra, Geometry and Number Theory study path
- Mathematics seminars
- Mathematics electives (MasterMath and Local courses)
- Mathematics specialisation Pure Mathematics compulsory courses
- MAT Probability and Statistics study path

Mathematics Analysis and Dynamical Systems study path

Programme Sections

- Mathematics Analysis and Dynamical Systems compulsory courses
- Mathematics Analysis and Dynamical Systems suggested electives

Mathematics Analysis and Dynamical Systems compulsory courses

Description

In the study path Analysis and Dynamical Systems, the following courses are compulsory.

Courses

Course Name	Period	Credits	Code
Dynamical Systems	P1+2+3	8EC	XMM_400429
Functional Analysis	P1+2+3	8EC	XMM_400328

Course Name	Period	Credits	Code
Partial Differential Equations	P1+2+3	8EC	XMM_400330

Mathematics Analysis and Dynamical Systems suggested electives

Description

The courses listed below are natural choices of electives for the study path Analysis and Dynamical Systems. Among these, the courses Bifurcation Theory, Optimal Transport, Numerical Methods for PDEs, Differential Geometry, Symplectic Geometry, Mathematical Biology, Applied Analysis: Financial Mathematics (MSc) are well connected to the research at the VU Mathematics department and will prepare you particularly well for your Master Project. This is a non-binding recommendation.

Courses

Course Name	Period	Credits	Code
Differential Geometry	P1+2+3	8EC	XMM_400509
Mathematical Biology	P1+2+3	8EC	XMM_400504
Numerical Methods for PDEs	P1+2+3	8EC	XMM_0072
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Bifurcation Theory	P4+5+6	8EC	XMM_0078
Interacting Particle Systems in the Lattice and Continuum	P4+5+6	8EC	XMM_0080
Inverse Problems in Imaging	P4+5+6	6EC	XMM_0047
Numerical Methods for Stochastic Differential Equations	P4+5+6	8EC	XMM_0071
Optimal Transport	P4+5+6	8EC	XMM_0063
Riemann Surfaces	P4+5+6	8EC	XMM_400325
Symplectic Geometry	P4+5+6	8EC	XMM_418036

Mathematics Algebra, Geometry and Number Theory study path

Programme Sections

- Mathematics Algebra, Geometry and Number Theory constrained choice
- Mathematics Algebra, Geometry and Number Theory suggested electives

Mathematics Algebra, Geometry and Number Theory constrained choice

Description

Constrained choice for the study path Algebra, Geometry and Number Theory: choose at least three courses from this list.

Courses

Course Name	Period	Credits	Code
Algebraic Geometry 1	P1+2+3	8EC	XMM_40001
Algebraic Number Theory	P1+2+3	8EC	XMM_400324
Algebraic Topology 1	P1+2+3	8EC	XMM_400600
Differential Geometry	P1+2+3	8EC	XMM_400509
Elliptic Curves	P4+5+6	8EC	XMM_400505

Mathematics Algebra, Geometry and Number Theory suggested electives

Description

The courses listed below are natural choices of electives for the study path Algebra, Geometry and Number Theory. Among these, the courses Analytic Number Theory, Algebraic Topology II, Algebraic Geometry II, Symplectic Geometry, Formal Methods in Mathematics, Modern Cryptography are well connected to the research at the VU Mathematics department and will prepare you particularly well for your Master Project. This is a non-binding recommendation.

Courses

Course Name	Period	Credits	Code
Advanced Algebraic Geometry	P1+2+3	8EC	XMM_418083
Analytic Number Theory	P1+2+3	8EC	XMM_0153
Commutative Algebra	P1+2+3	8EC	XMM_418136
Modern Cryptography	P1+2+3	8EC	XMM_0053
Algebraic Geometry 2	P4+5+6	8EC	XMM_40002
Algebraic Topology 2	P4+5+6	8EC	XMM_0017
Coding Theory	P4+5+6	8EC	XMM_418160
Formal Methods in Mathematics	P4+5+6	8EC	XMM_0066
Riemann Surfaces	P4+5+6	8EC	XMM_400325
Selected Areas in Cryptology	P4+5+6	8EC	XMM_40011
Symplectic Geometry	P4+5+6	8EC	XMM_418036
Topological Data Analysis	P4+5+6	8EC	XMM_0036

Mathematics seminars

Description

From 2025-26 on, the programme will offer 8 seminars on an alternate basis. Those students who have started in 2025-26, must complete 2 or 3 out of these 8 seminars.*

In 2025-26 and every other year after that, the 4 seminars linked under this tab will be taught.

For 2026-27 and every other year after that, the following 4 seminars are tentatively planned:

- Analysis of Neuroimaging Data,
- Mathematical Optimization,
- Algebraic Topology and Dynamical Systems,
- Concentration Inequalities.

As a non-binding recommendation, the following seminar choices are especially suitable for the study path

- Biomedical Mathematics: Network Dynamics, Symmetry and Synchrony, Sequential Monte Carlo Methods, Uncertainty Quantification, Analysis of Neuroimaging Data;
- Data Science and Optimization: Sequential Monte Carlo Methods, Uncertainty Quantification, Mathematical Optimization, Concentration Inequalities;
- Analysis and Dynamical Systems: Network Dynamics, Symmetry and Synchrony, Uncertainty Quantification, Algebraic Topology and Dynamical Systems, Concentration Inequalities;
- Algebra, Geometry, and Number Theory: p-adic Numbers and Their Applications, Algebraic Topology and Dynamical Systems;
- Probability and Statistics: Sequential Monte Carlo Methods, Uncertainty Quantification, Analysis of Neuroimaging Data, Concentration Inequalities.

* Those students who have started before or in 2024-25 and have completed 1 or 2 of the discontinued

- XM_41013 Master Seminar in Analysis and Dynamical Systems,
- XM_0058 Master Seminar in Algebra and Geometry,
- XM_41011 Master Seminar in Stochastics,

are exempt from the requirement to follow any of the 8 new seminars. However, if they wish to, they are allowed to choose a 2nd and/or 3rd seminar among the 8 new ones except (the upcoming in 2026-27) Algebraic Topology and Dynamical Systems will not be allowed in combination with (the discontinued) XM_41013 Master Seminar in Analysis and Dynamical Systems, due to a considerable overlap.

Courses

Course Name	Period	Credits	Code
p-Adic Numbers and Their Applications	P1+2	6EC	XM_0184
Uncertainty Quantification	P1+2	6EC	XM_0183
Network Dynamics, Symmetry and Synchrony	P4+5	6EC	XM_0181
Sequential Monte Carlo Methods	P4+5	6EC	XM_0182

Mathematics electives (MasterMath and Local courses)

Description

Without prior consent from the Examination Board, the student can take any of the (level-400 or higher) mathematics electives linked under this tab.

* XMM_0072 Numerical Methods for PDEs is not intended for students who have previously followed XMM_0028 Numerical Methods for Time-dependent PDEs and/or XMM_0046 Numerical Methods for Stationary PDEs, due to a considerable overlap.

** XMM_0080 Interacting Particle Systems in the Lattice and Continuum is not intended for students who have previously followed XMM_0064 Statistical Mechanics, due to a considerable overlap.

*** XM_0140 Applied Analysis: Financial Mathematics (MSc) is not intended for students who have previously followed X_400076 Applied Analysis: Financial Mathematics, due to a considerable overlap.

Additionally, without prior consent from the Examination Board but subject to the constraint of *at most two level-300 courses in all study paths of the programme except for the Teachers study path*, the student can take any of the following level-300 mathematics electives (to view the studyguides of these courses, search by course code or name on studiegids.vu.nl):

- X_400631 Differential Geometry (BSc),
- XB_0068 Differential Topology,
- XB_0079 Functional Analysis,
- XB_0134 Galois Theory,
- XB_0135 Markov Chains,
- XB_0136 Mathematical Statistics 2,
- X_401028 Measure Theory,
- XB_0098 Non-linear Dynamical Systems****,
- X_400632 Number Theory,
- X_400163 Partial Differential Equations (BSc).

**** is not intended for students who have previously followed the discontinued XM_0169 Non-linear Dynamical Systems (MSc), due to a considerable overlap.

Finally, note that, in all study paths of the programme except for the Teachers study path, the student can choose *up to 12 EC* of electives outside of the courses linked/listed here. These can be courses outside of mathematics, e.g. in literature. In the study paths in Applied/Pure Mathematics, these must be courses of *level 400 or higher*. In the Education study path, these may also be the following courses from the Beta4all/Wisk4all programme: Meetkunde (6 EC), Geschiedenis (6 EC). Advance permission must be obtained in writing from the Examination Board.

Courses

Course Name	Period	Credits	Code
Advanced Machine Learning	P1	6EC	XM_0010
Applied Stochastic Modeling	P1+2	6EC	X_400392
Statistical Models	P1+2	6EC	X_400418
Stochastic Processes for Finance	P1+2	6EC	X_400352
Advanced Algebraic Geometry	P1+2+3	8EC	XMM_418083
Algebraic Geometry 1	P1+2+3	8EC	XMM_40001
Algebraic Number Theory	P1+2+3	8EC	XMM_400324
Algebraic Topology 1	P1+2+3	8EC	XMM_400600
Analytic Number Theory	P1+2+3	8EC	XMM_0153
Asymptotic Statistics	P1+2+3	8EC	XMM_400323
Category Theory	P1+2+3	8EC	XMM_418114
Commutative Algebra	P1+2+3	8EC	XMM_418136
Complexity Theory	P1+2+3	8EC	XMM_0006
Continuous Optimization	P1+2+3	6EC	XMM_400446
Differential Geometry	P1+2+3	8EC	XMM_400509
Discrete Optimization	P1+2+3	6EC	XMM_400445
Dynamical Systems	P1+2+3	8EC	XMM_400429
Forensic Probability and Statistics	P1+2+3	8EC	XMM_0005
Functional Analysis	P1+2+3	8EC	XMM_400328
High-dimensional Probability	P1+2+3	8EC	XMM_0079
Infinity-categories	P1+2+3	8EC	XMM_0062
Lie Groups	P1+2+3	8EC	XMM_400350

Course Name	Period	Credits	Code
Mathematical Biology	P1+2+3	8EC	XMM_400504
Measure Theoretical Probability	P1+2+3	8EC	XMM_400244
Modern Cryptography	P1+2+3	8EC	XMM_0053
Modular Forms	P1+2+3	8EC	XMM_400599
Multivariate Statistics	P1+2+3	8EC	XMM_0081
Numerical Linear Algebra	P1+2+3	8EC	XMM_400329
Numerical Methods for PDEs	P1+2+3	8EC	XMM_0072
Partial Differential Equations	P1+2+3	8EC	XMM_400330
Quantum Computing	P1+2+3	8EC	XMM_0013
Quantum Information Theory	P1+2+3	8EC	XMM_0024
Systems and Control	P1+2+3	6EC	XMM_400332
Algorithms in Sequence Analysis	P2	6EC	X_405050
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Mathematical Optimization	P2	6EC	XM_0051
Performance of Networked Systems	P2	6EC	X_405105
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Optimization of Business Processes	P4+5	6EC	X_400422
Additive Combinatorics	P4+5+6	8EC	XMM_0025
Advanced Discrete Optimisation	P4+5+6	6EC	XMM_0075
Advanced Numerical Linear Algebra with Applications	P4+5+6	8EC	XMM_0076
Advanced Representation Theory and Applications	P4+5+6	8EC	XMM_0077
Algebraic Geometry 2	P4+5+6	8EC	XMM_40002
Algebraic Topology 2	P4+5+6	8EC	XMM_0017
Bayesian Statistics	P4+5+6	8EC	XMM_40004
Bifurcation Theory	P4+5+6	8EC	XMM_0078
Causality	P4+5+6	8EC	XMM_0057
Coding Theory	P4+5+6	8EC	XMM_418160
Computational Finance	P4+5+6	8EC	XMM_0060
Elliptic Curves	P4+5+6	8EC	XMM_400505
Formal Methods in Mathematics	P4+5+6	8EC	XMM_0066
Graph Symmetries and Combinatorial Designs	P4+5+6	8EC	XMM_0044
Homotopy Type Theory	P4+5+6	8EC	XMM_0061
Interacting Particle Systems in the Lattice and Continuum	P4+5+6	8EC	XMM_0080
Inverse Problems in Imaging	P4+5+6	6EC	XMM_0047
Lie Algebras	P4+5+6	8EC	XMM_0037
Machine Learning Theory	P4+5+6	8EC	XMM_0002
Numerical Methods for Stochastic Differential Equations	P4+5+6	8EC	XMM_0071
Operator Algebras	P4+5+6	8EC	XMM_418062
Optimal Transport	P4+5+6	8EC	XMM_0063
Queueing Theory	P4+5+6	6EC	XMM_400397
Riemann Surfaces	P4+5+6	8EC	XMM_400325
Scheduling	P4+5+6	6EC	XMM_400396
Selected Areas in Cryptology	P4+5+6	8EC	XMM_40011
Semidefinite Optimization	P4+5+6	8EC	XMM_0015
Set Theory	P4+5+6	8EC	XMM_418035
Stochastic Integration	P4+5+6	8EC	XMM_0049
Stochastic Processes	P4+5+6	8EC	XMM_400339
Symplectic Geometry	P4+5+6	8EC	XMM_418036
Topological Data Analysis	P4+5+6	8EC	XMM_0036

Mathematics specialisation Pure Mathematics compulsory courses

Description

The following components are compulsory in all three study paths of the specialisation Pure Mathematics.

Courses

Course Name	Period	Credits	Code
Introduction to Master Mathematics	Ac. Year (September)	6EC	XM_0095
Master Project Mathematics	Ac. Year (September)	36EC	X_400355

MAT Probability and Statistics study path

Programme Sections

- Mathematics Probability and Statistics compulsory courses
- Mathematics Probability and Statistics suggested electives

Mathematics Probability and Statistics compulsory courses

Description

In the study path Probability and Statistics, the following courses are compulsory.

Courses

Course Name	Period	Credits	Code
Asymptotic Statistics	P1+2+3	8EC	XMM_400323
Measure Theoretical Probability	P1+2+3	8EC	XMM_400244
Stochastic Processes	P4+5+6	8EC	XMM_400339

Mathematics Probability and Statistics suggested electives

Description

The following are natural choices of electives for the study path Probability and Statistics. The majority of the courses on this list, except for Causality, are well connected to the research at the VU Mathematics department and will prepare you particularly well for your Master Project. This is a non-binding recommendation.

Courses

Course Name	Period	Credits	Code
Advanced Machine Learning	P1	6EC	XM_0010
Applied Stochastic Modeling	P1+2	6EC	X_400392
Statistical Models	P1+2	6EC	X_400418
Stochastic Processes for Finance	P1+2	6EC	X_400352
Forensic Probability and Statistics	P1+2+3	8EC	XMM_0005
Mathematical Biology	P1+2+3	8EC	XMM_400504
Optimization of Business Processes	P4+5	6EC	X_400422
Bayesian Statistics	P4+5+6	8EC	XMM_40004
Causality	P4+5+6	8EC	XMM_0057
Machine Learning Theory	P4+5+6	8EC	XMM_0002
Queueing Theory	P4+5+6	6EC	XMM_400397
Stochastic Integration	P4+5+6	8EC	XMM_0049

Courses

Additive Combinatorics

Course Code	XMM_0025
Credits	8
Period	P4+5+6

Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

-

Method of Assessment

-

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Advanced Algebraic Geometry

Course Code	XMM_418083
Credits	8
Period	P1+2+3
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Advanced Discrete Optimisation

Course Code	XMM_0075
Credits	6
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

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Advanced Machine Learning

Course Code	XM_0010
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai

Examiner	prof. dr. S. Bhulai
Teaching Staff	prof. dr. S. Bhulai
Teaching method(s)	Lecture

Course Objective

There are several learning objectives for this course. After completion of this course, the student should be able to:

- understand the capabilities and the limitations of machine learning,
- implement machine learning algorithms in Python,
- know relevant machine learning algorithms for both supervised and unsupervised learning problems,
- select the right machine learning models for real-world use cases,
- understand when to apply online learning, reinforcement learning, and deep learning,
- interpret the outcomes of machine learning algorithms.

Course Content

Machine learning is the science of getting computers to act without being explicitly programmed. It is so pervasive today that people use it in everyday life without knowing it. In this course, you will learn about the most effective machine learning techniques and gain practice implementing them and getting them to work yourself. We will discuss the theoretical underpinnings as well as the practical know-how needed to apply these techniques to new problems.

Additional Information Teaching Methods

Lectures (14 x 2 hours), including guest speakers.

Method of Assessment

Written exam.

Literature

Slides and additional material that will be posted on Canvas.

Additional Information Target Audience

mBA, mBA-D, mMath, mCS

Recommended background knowledge

The VU course Linear Algebra and the VU course Statistics, or equivalent courses, are preferred.

Advanced Numerical Linear Algebra with Applications

Course Code	XMM_0076
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and

course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

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Advanced Representation Theory and Applications

Course Code	XMM_0077
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

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AI in het Onderwijs

Course Code	O_MAIIHO
Credits	3
Period	P1, P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. ir. F.F.J. Hermans
Examiner	dr. H.B. Westbroek
Teaching Staff	prof. dr. ir. F.F.J. Hermans, dr. S.M. Calor
Teaching method(s)	Seminar, Lecture

Course Objective

Het doel van dit vak is op basis van kennis over AI en over onderwijs theorieën een helder beeld schetsen van hoe je je als docent en school wilt verhouden tot generatieve AG.

Course Content

Studenten leren in dit vak:

- Hoe generatieve AI werkt
- Wat de impact van AI op onderwijs kan zijn, vanuit een aantal verschillende theorieën en invalshoeken
- Wat de risico's van AI zijn, bijvoorbeeld mbt bias en deskilling
- Reflecteren op het gebruik van AI in de eigen onderwijspraktijk

Belangrijk: Voor dit vak is geen technische voorkennis vereist en het is voor alle studenten van de lerarenopleiding (en aanverwante studies) geschikt, zeker niet louter voor de beta's!

Additional Information Teaching Methods

Het vak bestaat uit:

- 5 verplichte hoorcolleges (10 uur)
- Huiswerkopdrachten maken (25 uur)
- Lezen literatuur (25 uur)
- Uitwerken beleidsplan (24 uur)

Method of Assessment

Studenten sluiten dit vak af door het schrijven van een beleidsplan over het gebruik van AI binnen hun vak of school.

Literature

Zie Canvas

Algebraic Geometry 1

Course Code	XMM_40001
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Algebraic Geometry 2

Course Code	XMM_40002
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Algebraic Number Theory

Course Code	XMM_400324
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Algebraic Topology 1

Course Code	XMM_400600
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Seminar, Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Algebraic Topology 2

Course Code	XMM_0017
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Algorithms in Sequence Analysis

Course Code	X_405050
Credits	6

Period	P2
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	R. Kumar
Examiner	dr. S. Pissis
Teaching Staff	dr. J.M. Urtasun Elizari, dr. ir. K.A. Feenstra, dr. S. Pissis, prof. dr. J. Heringa, dr. Z. Işik MSc, H. Verbeek MSc
Teaching method(s)	Computer lab, Lecture

Course Objective

Have you ever wondered how we can track a gene across 3 billion years of evolution? Or how you can use the genome information of a given cancer patient to find out what may be wrong? Sequence alignment can be used to compare genomes, genes or proteins from bacteria all the way to humans, while further derived algorithms may be employed to make a phylogeny (to find out about evolutionary relationships), find a functional motif in a protein sequence, or a viral sequence in a genome. In this course we focus on the most important algorithms for biological sequence analysis that can be applied to real scientific problems in biology.

After completing the course successfully,

- You can demonstrate in-depth knowledge about the theory of sequence analysis methods.
- You know how to implement several of the most important sequence alignment and analysis algorithms.
- You understand the mathematical formulation behind sequence analysis algorithms.
- You will have understanding and skills to apply sequence analysis algorithms to protein and DNA sequences.
- You will have hands-on experience in tackling biological problems using sequence analysis algorithms, including the statistical framework of HMMs and algorithms used in genome sequencing and analysis.
- You can decide which algorithm is best suited for a particular biological sequence analysis problem.
- You can read through scientific and technical literature and learn to translate algorithms described in text and formal mathematical notation into computer code.

Course Content

Theory:

- Dynamic programming
- database searching
- pairwise and multiple alignment
- probabilistic methods including HMMs
- pattern matching
- entropy measures
- evolutionary models
- and phylogeny

Practical:

- Programming (in Python) an alignment algorithm based on dynamic programming;
- Aligning sequencing data from tumors to the human genome and analysing structural variants;
- Programming (in Python) an implementation of HMMs and using it to predict protein domain structure.

Additional Information Teaching Methods

13 lectures: 2 two-hour lectures per week.
13 computer practicals and associated assignments: 2 two-hour hands-on sessions per week.

Method of Assessment

The final grade for this course will consist of 50% practical work (see above) and 50% theoretical assessment. (assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

The theoretical assessment will be an oral and/or written exam (depending on number of students).

Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Entry Requirements

Bachelor in any science discipline (including medicine).
Basic programming skills (Python) and an interest in biological problems.

Literature

Course material on Canvas.

Books: Durbin, R., Eddy, S.R., Krogh, A., Mitchison, G.. Biological Sequence Analysis. Cambridge University Press, 1998, 350 pp., ISBN 0521629713.

Recommended reading: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics. Garland Science 2008. ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Artificial Intelligence
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)

We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. Ideally, this laptop supports a command line shell, Python 3 and a text editor with syntax highlighting -- either standalone (e.g. Atom or Sublime Text) or as part of a simple IDE (e.g. Spyder). As such, we recommend the Anaconda python distribution regardless of operating system, along with PuTTy or PowerShell for Windows users specifically.

The course is taught in English.

Analytic Number Theory

Course Code	XMM_0153
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

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Applied Analysis: Financial Mathematics (MSc)

Course Code	XM_0140
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. A.C.M. Ran
Examiner	prof. dr. A.C.M. Ran
Teaching Staff	prof. dr. A.C.M. Ran
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

This course is taught jointly with X_400076 Applied Analysis: Financial Mathematics for BSc Mathematics, see the studyguide of X_400076.

Course Content

See the studyguide of X_400076

Additional Information Teaching Methods

See the studyguide of X_400076

Method of Assessment

See the studyguide of X_400076

Literature

See the studyguide of X_400076

Additional Information Target Audience

MSc Business Analytics, MSc Mathematics

Note that this course is not intended for students who followed X_400076 in their Bachelor's, due to considerable overlap.

Recommended background knowledge

See the studyguide of X_400076

Applied Stochastic Modeling

Course Code	X_400392
Credits	6
Period	P1+2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R. Bekker
Examiner	dr. R. Bekker
Teaching Staff	dr. R. Bekker

Teaching method(s)	Seminar, Lecture

Course Objective

During this course you will get acquainted with the most often used stochastic models and how they are applied in practice. The emphasis is on the variety of stochastic models (and their analysis) that appear in practice, rather than an in-depth study of a single-class of models. During the course you learn to handle such practically motivated problems as an independent researcher; this means that you:

- are able to determine the appropriate model
- can formulate the problem mathematically correct
- are able to solve the stochastic model
- know how to interpret the outcome.

Course Content

This course deals with a number of stochastic modeling techniques that are often used in practice. They are motivated by showing the business context in which they are used. Topics we deal with are: time-dependent Poisson processes and infinite-server queues, renewal processes and simulation, birth-death-processes, basic queueing models, and inventory models. We also repeat and extend certain parts of probability theory.

Additional Information Teaching Methods

Lecture and instruction.

Method of Assessment

Written examination and two hand-in assignments (one in each period).
The assignment in period 1 is graded as a pass-fail, and the assignment in period 2 counts for 20%; the written examination counts for 80%. The weighted average grade should be sufficient, but for the written examination there is a minimum requirement of a 5.0. The same grading scheme applies to the resit, unless the grade for the resit exam is higher than the grade for the handin-assignment in period 2, in which case the grade for the resit is determined entirely by the resit exam.
There is no resit for the hand-in assignment in period 2. The hand-in assignment in period 1 may be repaired in case of a fail.

Literature

- Lecture notes of Ger Koole (link will be made available via Canvas).
- Recommended: H.C. Tijms, A First Course in Stochastic Models, 2003. This is available as e-book via the VU library (ubvu), free of charge.
- Two scientific paper(s) will be made available via Canvas.

Additional Information Target Audience

mBA, mMath

Recommended background knowledge

Probability theory, Poisson process, Markov chains in continuous time

Asymptotic Statistics

Course Code	XMM_400323
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see
https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Basic Models of Biological Networks

Course Code	X_418154
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. F.J. Bruggeman
Examiner	prof. dr. F.J. Bruggeman
Teaching Staff	prof. dr. F.J. Bruggeman, dr. G. Stolf Jeuken
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

This course focuses on teaching the creation and analysis of mathematical models for biological systems across various levels of organization, from molecular circuits to cellular systems and potentially ecosystems. The primary modelling formalism explored is ordinary differential equations (ODEs), providing a foundation for understanding biological dynamics. The course aims to equip students with the ability to reproduce models from scientific literature and apply them to solve practical problems. It combines lectures with workgroups involving problem-solving using both computational tools (Python) and pen and paper.

Learning Goals:

- Learn to develop and analyse mathematical models (primarily ODE-based) for diverse biological networks (molecular, cellular, etc.).
- Understand how to represent biological processes, particularly molecular mechanisms like enzyme kinetics and protein regulation, using mathematical formalisms like state-transition diagrams and mass-action kinetics.
- Gain proficiency in simplifying complex mechanisms into net process models suitable for integration into larger system models.
- Develop the skills to reproduce existing models from scientific papers and apply modelling techniques to solve real problems.
- Acquire the intuition necessary to construct useful and informative models by analysing various examples.
- Understand the difference between mechanistic and phenomenological/empirical models and their applications.

Course Content

The course slowly builds on the principles required to make useful models of biological dynamics, starting from first principles:

(1) Fundamentals of Dynamical Systems in Biology:

- Introduction to change, dynamics, systems organization, and the role of energy in biological systems.

(2) Modelling Molecular Circuits:

- Using state-transition diagrams to visualize molecular states and transitions.
- Applying mass-action kinetics to describe the rates of spontaneous molecular reactions and processes within state-transition diagrams.
- Developing mathematical models based on state-transition diagrams.

(3) Net Models and Simplification:

- Modelling enzyme kinetics, including catalytic cycles, rapid-equilibrium approximations, steady-state methods, and cooperative effects.
- Developing equilibrium-binding equations for regulatory proteins (e.g., transcription factors, receptors).
- Reducing complex state-transition diagrams to simpler net process models for use in larger systems.

(4) Models of Larger Systems:

- Applying simplified models to study metabolism, signalling, and signal transduction.
- Introduction to phenomenological and empirical models for systems where detailed mechanisms are unknown or too complex (e.g., cell growth, ecosystems).

(5) Practical Application:

- Solving exercises using Python and pen and paper to reinforce concepts and build modelling skills.

Additional Information Teaching Methods

- Lectures
- Workgroups
- Self-study
- Computer practicals
- Computer modelling tutorials

Method of Assessment

- 80% Written exam
- 20% Computer exam (Python)

You must have a grade higher than 5.5/10 on both exams to pass.

Literature

A course syllabus written by the lecturer

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Mathematics

Bayesian Statistics

Course Code	XMM_40004
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	

Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

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Bifurcation Theory

Course Code	XMM_0078
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

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Bioinformatics for Translational Medicine

Course Code	X_405092
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. E. Bosdriesz
Teaching Staff	dr. E. Bosdriesz, dr. J.M. Urtasun Elizari
Teaching method(s)	Practical, Lecture

Course Objective

Observations from biological high-throughput experiments allow us to improve diagnosis and give a personalized treatment plan for patients. However, integrating data from several sources and using this data for predictions is non-trivial.

This is a theoretical and practical Bioinformatics course on computational methods for Translational Medicine; we will focus on Bioinformatics methods that are used to predict the clinical outcome for patients and analysis methods to obtain deeper understanding of complex diseases, by combining data from various high-throughput experiments such as proteomics, microarrays and next-generation sequencing as well as existing biological databases.

After completing the course successfully,

- You can interpret research results in the context of biology
- You will be aware of Bioinformatics methods that are applicable to the area of Translational Medicine
- You can combine these methods into a creative solution using large-scale biological experiments
- You will have hands-on experience in handling large biological datasets, and will understand the complexity of the biological data both from high-throughput experiments and existing biological databases
- You will be able to work as a team to execute a large, complex biomedical bioinformatics project, and report on the results
- You will be familiar with a few in-depth research topics that lie within the expertise area of several (Bioinformatics) researchers, among others from the VU, UvA, AUMC and NKI

Course Content

Theory

- Computational analysis of molecular profiling techniques, such as WGS, WES, proteomics, RNA sequencing, and arrayCGH.
- Computational methods applied to these data types, such as: machine learning, normalisation, regularization, feature selection, classification, read mapping, clustering.
- All data analysis is relevant in a clinical setting, for diagnosis, treatment decisions or biomarker discovery.

Practical

- A large assignment for which you have to build a classifier based on molecular profiling data (the precise data-set is to be determined). You need to hand in predictions, write a paper and give a presentation. Note that this is a group project, and to participate in this practical you need to have obtained at least 12 ECT from Master-level courses.
- A smaller assignment that involves analysing and understanding molecular profiling data of another modality than the large classifier assignment.

Additional Information Teaching Methods

- 13 lectures (2 two-hour lectures per week)
- 12 computer practicals (2 two-hour sessions per week)

Method of Assessment

The final grade for this course will consist of 50% practical work (see above) and 50% theoretical assessment.

Practical assessment (50%):

- Classifier assignment (45%)
- Other data modality assignment (5%)

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Theoretical assessment (50%):

- Oral or written exam (depending on number of course students)
- The exam is based on a selection of 8-10 scientific papers in the field of Bioinformatics & Translational Medicine.

Entry Requirements

Importantly: to participate in the group assignment, you need to have obtained a minimum of 12 ECT of Master level courses. In addition, some basic programming skills (either in R or Python) are required, as well as some basic knowledge on molecular biology. If you are not following the Master Bioinformatics and Systems Biology, it is advisable to first follow the Master course "Fundamentals of Bioinformatics".

Literature

- Course material on Canvas, including slides and recordings of lectures.
- 8-10 scientific papers are provided, and make up the course syllabus.

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

The course is taught in English.

- Compulsory course for students in Master Bioinformatics and Systems Biology
- Optional course for students with a Bachelor in Physics, Chemistry, Mathematics, Computer Science, Biology, or Biomedical Sciences (see requirements below).

Recommended background knowledge

If you are not following the Master Bioinformatics and Systems Biology, it is advisable to first follow the Master course "Fundamentals of Bioinformatics".

Brain Imaging

Course Code	P_MBRIMAG_AI
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. T.H.J. Knapen
Examiner	dr. T.H.J. Knapen
Teaching Staff	dr. T.H.J. Knapen, mr. S. Zhang MSc, mr. R. van de Klundert MSc
Teaching method(s)	Seminar, Lecture

Course Objective

fMRI is the main method we have of imaging activations of the living human brain, and thereby probe the computational mechanisms underpinning our cognitive repertoire. The goal of this course is to give you an idea of the awesome possibilities and annoying pitfalls of functional brain imaging, but most importantly, to give you a solid foundation for further learning. At the end of the course, you will be able to devise a valid fMRI experiment, and perform the basic analysis on the resulting data using state-of-the-art open science and open source tools.

Course Content

In this course we will teach you the ins and outs of brain imaging, that is, fMRI. We will teach you everything from the basics of signal analysis, to experimental design, to statistics. Some of the newest cutting-edge techniques, including pattern classification analysis, connectivity modelling, and resting state network analysis, are also discussed.

Additional Information Teaching Methods

Every week, there will be one or two lectures, interspersed with 3 to 4 practicals/'werkcolleges' in total. The course is broadly divided into two parts; the first half of the course serves to teach you the very basics of signal analysis and experimentation. We believe this basis is necessary to later start to think independently and academically about research in your future field. In this first phase of the course the weekly lecture will treat theory while the practicals will allow you to wet your toes with this material. This way we try to combine theory and practice.

In the second half of the course, you will already know a lot about what Brain Imaging entails. Then, we will switch gears a bit, and teach you what's going on in the neuroimaging field right now. That means that during the weekly lecture we will use research articles to illustrate the state of the art. In the practicals we'll move towards letting you perform an entire fMRI analysis yourselves. In this second part of the course we'll also focus more and more on recent articles that show us the state-of-the-art in neuroimaging.

Method of Assessment

Final Exam, open-end & MC questions 70%
Quizzes on Canvas 20%
Perusall reading participation 10%

Entry Requirements

Fluency in the Python programming language.

Literature

We will use articles describing current research.

Additional Information

Prior knowledge of Python programming and statistics is important for following the practicals.

Category Theory

Course Code	XMM_418114
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See https://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See https://elo.mastermath.nl

Method of Assessment

See https://elo.mastermath.nl

Additional Information Target Audience

mMATH

Custom Course Registration

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Causality

Course Code	XMM_0057
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Coding Theory

Course Code	XMM_418160
Credits	8

Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

-

Method of Assessment

-

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Commutative Algebra

Course Code	XMM_418136
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

-

Method of Assessment

-

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Complex Analysis

Course Code	X_400386
Credits	6
Period	P4+5
Course Level	300
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I. Klang
Examiner	dr. I. Klang
Teaching Staff	dr. I. Klang
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

- The student can decide whether a complex function is analytic (=differentiable in the complex sense) and knows the connection with the Cauchy-Riemann equations.
- They can do computations with elementary functions such as exp/log/sin/cos over the complex numberts.
- They can integrate analytic functions along a path on the complex plane, using the theorem of Cauchy-Goursat and its corollaries.
- They can compute Laurent series and determine the type of singularities of analytic functions.
- They can compute integrals of complex functions using the residue theorem and know how to use this to compute integrals of real functions.

Course Content

In complex analysis one generalizes the standard concepts of real analysis such as differentiation and integration from the real line to the complex plane. Although these generalizations arise very naturally and all standard examples of functions are also differentiable in the complex sense, the latter property surprisingly turns out to be much stronger. As a consequence, complex differentiable functions immediately obey very special properties which we are going to explore in this course. In particular, they lead to completely new and efficient methods for computing integrals of real functions.

During the lectures the following topics will be treated:

- complex differentiation and the Cauchy-Riemann equations
- complex integration and the theorem of Cauchy-Goursat
- elementary properties of complex differentiable functions
- singularities, Laurent series and the residue theorem
- application to integrals of real functions

Additional Information Teaching Methods

Lecture (2 hours) and tutorial class (2 hours)

Method of Assessment

Two written exams (40%+40%) and two hand-in homeworks (10%+10%). The retake exam counts for 100% of the final grade.

Literature

Churchill, R. V., & Brown, J. W.: Complex variables and applications. Ninth edition, 2014, McGraw-Hill Book Co., New York

Additional Information Target Audience

Bachelor Mathematics Year 2

Recommended background knowledge

Calculus, Analysis, Linear algebra are necessary background.

Complexity Theory

Course Code	XMM_0006
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath, mSFM

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Computational Finance

Course Code	XMM_0060
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai

Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath, mSFM

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Continuous Optimization

Course Code	XMM_400446
Credits	6
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

-

Additional Information Target Audience

mMath; mBA

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Didactics 1

Course Code	O_MLDIDAC_1F
Credits	6
Period	P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	prof. dr. M. Meeter, dr. I. Pauw, dr. S. Donszelmann, D.S. Hoefakker-Laanstra, drs. M.A. Huijskes MA, dr. H.B. Westbroek, dr. C.D.P. van Oeveren, drs. A.J.C. Monquil, dr. O.P.M. van Buuren, drs. Y.G. Meindersma, C.L. Geraedts, dr. B. de Vries, drs. R. Cornelisse MSc, M. Beukering, drs. E.D. van Noort, drs. H.R. Goudsmit, drs. M.M. Bout, prof. dr. ir. F.F.J. Hermans, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, N. el Abdellaoui MSc, drs. J.K.W. Riksen, drs. N.H. Ypenburg
Teaching method(s)	Seminar, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Course Content

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio’s vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio’s werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici,

pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Didactics 1

Course Code	O_MLDIDAC_1
Credits	6
Period	P1
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. L.J. van Well-van Grootheest, drs. M.A. Huijskes MA, drs. J.K.W. Riksen, dr. O.P.M. van Buuren, drs. E.D. van Noort, drs. Y.G. Meindersma, dr. S.M. Calor, dr. I. Pauw, drs. A.J.C. Monquil, M. Beukering, drs. N.H. Ypenburg, dr. C.D.P. van Oeveren, dr. S. Donszelmann, drs. M.M. Bout, drs. H.R. Goudsmit, dr. H.B. Westbroek, C.L. Geraedts, N. el Abdellaoui MSc, drs. R. Cornelisse MSc, D.S. Hoefakker-Laanstra, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, prof. dr. M. Meeter
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Course Content

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio's vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Interactieve hoorcolleges, werkcolleges en zelfstudiemodules.

Method of Assessment

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Didactiek 2 Semester 1

Course Code	O_MLDIDAC2
Credits	9
Period	P2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	C.L. Geraedts, prof. dr. M. Meeter, drs. M.M. Bout, drs. A.J.C. Monquil, drs. E.D. van Noort, drs. R. Cornelisse MSc, dr. H.B. Westbroek, dr. S. Donszelmann, dr. O.P.M. van Buuren, D.S. Hoefakker-Laanstra, N. el Abdellaoui MSc, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, drs. L.J. van Well-van Grootheest, drs. H.R. Goudsmit, drs. Y.G. Meindersma, M. Beukering, drs. M.A. Huijskes MA, drs. N.H. Ypenburg, dr. I. Pauw, dr. S.M. Calor
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Course Content

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 2 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Didactiek 2 is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Didactiek 2 staan de volgende kernpraktijken centraal: (1) Zien wie er in je klas zit, (2) Leerprocessen bevorderen, (3) Leerprocessen zichtbaar maken, (4) Leiding geven, (5) Een veilig leerklimaat creëren, en (6) Leerprocessen toetsen. Ten opzichte van de cursus Didactiek 1 wordt de focus verlegd van de (individuele) les naar het leerproces van de leerling. De reikwijdte van het didactisch denken en handelen wordt daarmee ook groter: er wordt een begin gemaakt met het ontwerpen en uitvoeren van reeksen van lessen.

Het fysieke onderwijs van Didactiek 2 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD in periode 2 genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. In periode 3 staan, na de peergroupsessies in de ochtend, een aantal grote onderwijsthema's centraal.

Het onderdeel AD in de cursus Didactiek 2 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Entry Requirements

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

NB: Voor studenten die vóór 1 september 2023 gestart zijn geldt in studiejaar 2023/2024 het curriculum van 2022/2023. Dit betekent dat het vak Didactiek 2 een studielast van 6 EC (ipv 9EC) heeft (en ze zich voor die variant intekenen). Studenten die de colleges van Didactiek 2 (6 EC) nog niet (geheel) gevolgd hebben sluiten aan bij colleges van het nieuwe vak Didactiek 2 (9 EC). Het is tot en met augustus 2024 nog mogelijk om Didactiek 2 (6EC) uit het oude curriculum af te ronden en het examen onder de eindtermen van het oude curriculum af te leggen.

Didactiek 2 Semester 2

Course Code	O_MLDIDAC2F
Credits	9
Period	P5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. Y.G. Meindersma, drs. E.D. van Noort, prof. dr. ir. F.F.J. Hermans, drs. N.H. Ypenburg, dr. O.P.M. van Buuren, dr. H.B. Westbroek, dr. I. Pauw, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, N. el Abdellaoui MSc, dr. B. de Vries, drs. A.J.C. Monquil, M. Beukering, C.L. Geraedts, drs. R. Cornelisse MSc, drs. H.R. Goudsmit, dr. C.D.P. van Oeveren, drs. M.A. Huijskes MA, D.S. Hoefakker-Laanstra, drs. J.K.W. Riksen, dr. S. Donszelmann, drs. M.M. Bout, prof. dr. M. Meeter
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Course Content

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 2 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Didactiek 2 is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Didactiek 2 staan de volgende kernpraktijken centraal: (1) Zien wie er in je klas zit, (2) Leerprocessen bevorderen, (3) Leerprocessen zichtbaar maken, (4) Leiding geven, (5) Een veilig leerklimaat creëren, en (6) Leerprocessen toetsen. Ten opzichte van de cursus Didactiek 1 wordt de focus verlegd van de (individuele) les naar het leerproces van de leerling. De reikwijdte van het didactisch denken en handelen wordt daarmee ook groter: er wordt een begin gemaakt met het ontwerpen en uitvoeren van reeksen van lessen.

Het fysieke onderwijs van Didactiek 2 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD in periode 2 genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. In periode 3 staan, na de peergroupsessies in de ochtend, een aantal grote onderwijsthema's centraal.

Het onderdeel AD in de cursus Didactiek 2 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Entry Requirements

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

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- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

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Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

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Differential Geometry

Course Code	XMM_400509
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

See https://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and

course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See https://elo.mastermath.nl

Method of Assessment

See https://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

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Discrete Optimization

Course Code	XMM_400445
Credits	6
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Dynamic Programming and Reinforcement Learning

Course Code	XM_0093

Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. G.M. Koole
Examiner	prof. dr. G.M. Koole
Teaching Staff	V.G.M. Francois Lavet, prof. dr. G.M. Koole
Teaching method(s)	Seminar, Lecture

Course Objective

Dynamic Programming (DP) and Reinforcement Learning (RL) are fields concerned with decision making over time. After completing this course, the student

- is familiar with the commonly used algorithms for solving dynamic optimization problems;
- understands the main features of these algorithms, their strengths and weaknesses including their convergence properties;
- can implement them in an appropriate language;
- can model real-world decision problems into a DP or RL framework and solve moderately sized problems;
- has knowledge of the historical development of DP and RL and has an idea of possible future developments.

Course Content

This course is concerned with reinforcement learning and its origin dynamic programming. These are fields dealing with goal-directed decision making over time, such as finding your way in an unknown area, playing a game or pricing airline tickets.
We look at these areas from different angles:

- we deal with full-information "planning" problems, but also with partial-information "learning" problems
- we consider different algorithms, some of which are guaranteed to find the best solution, but also heuristics
- we consider high-dimensional problems (such as games) and methods to solve them
- we look at small toy problems to understand algorithms and sharpen our intuition, but also bigger problems for which we learn how to implement algorithms (in python)
- we look at different types of applications, both from AI (search problems, games) and OR

Additional Information Teaching Methods

Lectures and practical work integrated

Method of Assessment

Programming exercises and final exam.
The 3 assignments each count for 10% and the exam for 70%.
The minimal passing grade for the exam is 5.0.

Literature

Slides and lecture notes

Additional Information Target Audience

mBA, mAI, mCS, mBa-D, mMath

Recommended background knowledge

Programming experience in python

Dynamical Systems

Course Code	XMM_400429
Credits	8
Period	P1+2+3
Course Level	400

Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Additional Information

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Dynamical Systems

Course Code	X_400637
Credits	6
Period	P4+5
Course Level	300
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Bick
Examiner	dr. C. Bick
Teaching Staff	dr. C. Bick
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

At the end of this course students ...

- have a basic understanding of discrete and continuous time dynamical systems;
- are introduced to the concepts of fixed points/equilibria and their stability;
- know the existence an uniqueness theorem for initial value problems;
- are able to solve constant coefficient linear (matrix) differential equations and know the elementary linear algebra needed for that;
- can analyze one- and two-dimensional dynamics and are able to draw phase space / phase plane pictures;
- know elementary bifurcation theory and in particular the saddle-node bifurcations;

- can recognize and analyze gradient, conservative and Hamiltonian dynamical systems;
- know and understand the stable and unstable manifold theorem and the principle of linearization;
- know what limit sets are;
- are able to apply the Poincaré-Bendixson theorem.

Course Content

This course provides an introduction to dynamical systems with a main focus on continuous time dynamical systems defined by ordinary differential equations. Subjects are:

1. Basic notions of dynamical systems theory in discrete and continuous time;
2. Existence an uniqueness of initial value problems;
3. Linear systems and elementary linear algebra;
4. One-dimensional dynamics and two-dimensional phase plane pictures;
5. Elementary bifurcation theory and saddle-node bifurcations;
6. Gradient and Hamiltonian dynamics;
7. Stable and unstable manifolds and linearization;
8. Limiting behavior;
9. The Poincaré-Bendixson theorem.

Additional Information Teaching Methods

Regular instruction class in combination with tutorial classes.

Method of Assessment

Midterm and final exams counting 50% each. The resit exam counts for 100%.

Literature

- Bob Devaney (1992). *A First Course in Chaotic Dynamical Systems: Theory and Experiment.* Westview Press.
- Lawrence Perko, "Differential Equations and Dynamical Systems", third edition, Springer-Verlag 2001, ISBN-10: 0387951164.
- Additional reading material and lecture notes will be communicated to the students via Canvas.

Additional Information Target Audience

Bachelor Mathematics Year two

Additional Information

Additional information on Canvas

Recommended background knowledge

First year courses Calculus and Analysis. Linear Algebra.

Elliptic Curves

Course Code	XMM_400505
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Entrepreneurship in Analytics and AI

Course Code	XM_0090
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. Stam
Examiner	prof. dr. W. Stam
Teaching Staff	prof. dr. W. Stam, H.V. Nguyen, drs. G.J. Beekman MA, P. Afonso Branco Ramos Pinto
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

Academic and research skills: Students are able to generate a promising new business idea in the field of AI & Analytics and develop a prototype that validates the proposed customer needs and market potential.

Bridging theory and practice: Students understand the core theories and concepts in the field of entrepreneurship, as reflected by their ability to analyze, compare, and apply these theories and concepts to analyze and solve the key management challenges faced by real-world entrepreneurs.

Entrepreneurial soft skills: Students are able to effectively work in a team for generating and developing a new venture idea and can pitch the business concept to external stakeholders in a professional, engaging manner.

Course Content

Entrepreneurship in Analytics and AI is an engaging course for students who like to learn about entrepreneurship, its role and importance in our society, and the process by which entrepreneurs transform new ideas into successful AI-driven business ventures.

In this course, students will learn the ropes of this emerging field via three educational pillars:

1. Students will learn to understand and apply theories from economics, sociology, and psychology to study key topics in entrepreneurship, including innovation, new venture creation, entrepreneurial competences, entrepreneurial finance, scaling and growth of new ventures, corporate entrepreneurship, and entrepreneurial ecosystems. Theoretical understanding of these subjects is applied to real world cases focusing on key management challenges faced by entrepreneurs and how to creatively overcome these challenges.
2. Students will familiarize with a set of entrepreneurial soft skills, which they will put into practice when they will be asked to approach different stakeholders to further their team projects to gain commitments or to obtain important information. For this purpose, the students will receive training in generating initial ideas for

opportunities as well as networking and network utilization.
3. Students will be introduced to the actions and strategies that promote the process of new venture creation. For this purpose, the students, supported by the lecturers and coaching, will write (and pitch) a Student Design Sprint report (including prototype) around an innovative new venture opportunity in the field of Analytics and/or AI that they deem interesting for entrance into the market.

Additional Information Teaching Methods

During this course, students will attend both (guest) lectures and tutorials. The lectures explain relevant concepts, theories, and frameworks in the field of entrepreneurship and illustrate their usefulness using various practical examples. The tutorials are designed to facilitate students in making progress with their team projects by providing appropriate guidance and feedback.

Method of Assessment

The final grade for this course is based on a written individual exam, several individual pass/fail assignments, and a Student Design Sprint (SDS) project assignment.

The written exam and pass/fail assignments are individual assessments whereas the SDS assignment is a team assessment. To pass the course, the exam and SDS assignment should be graded as sufficient. Both forms of assessment have one option for re-sit in case students received an insufficient score.

Literature

The literature for this course constitutes a selection of academic articles and case studies.

Forensic Probability and Statistics

Course Code	XMM_0005
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	prof. dr. K. Slooten
Teaching method(s)	Seminar, Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the

administration of your university.

Formal Methods in Mathematics

Course Code	XMM_0066
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

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Fourier Analysis

Course Code	XB_0005
Credits	6
Period	P1+2
Course Level	200
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R.C.A.M. van der Vorst
Examiner	prof. dr. R.C.A.M. van der Vorst
Teaching Staff	prof. dr. R.C.A.M. van der Vorst
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

At the end of this course the student is able to:

- Calculate the Fourier series of a given Riemann-integrable periodic function.
- Analyze the convergence properties of Fourier series.

- Apply the theory of good kernels in (theoretical) examples.
- Apply Plancherel's theorem and the convergence of Fourier series in examples.
- Prove theoretical statements about the Fourier series.
- Calculate the Fourier transform of a given function of moderate decrease.
- Use the Fourier transform to solve certain PDE's.
- Apply Plancherel's theorem , the Fourier inversion formula and the Poisson summation formula in examples.
- Analyze whether functions are Schwartz functions or functions of moderate decrease.
- Prove theoretical statements about the Fourier transform on the real line.

Course Content

Topics that will treated are:

- Uniform convergence
- The genesis of Fourier Analysis, in particular the investigation of the wave equation
- Basic Properties of Fourier Series (uniqueness, convolutions, Dirichlet and Poisson kernels)
- Convergence of Fourier Series (pointwise, mean-square)
- Cesàro and Abel Summability
- Some applications of Fourier Series
- The Fourier transform on the real line (definition, inversion, Plancherel formula)
- Applications of the Fourier transform to some partial differential equations

Additional Information Teaching Methods

Lectures (1x2 hours per week) and Tutorials (1x2 hours per week).
Active participation during the tutorials is expected!

Method of Assessment

There are homework assignments, a midterm and a final. The precise weight of these components towards the final grade will be communicated on canvas. There is one resit opportunity for the full course. The grade for the homework does not count toward the grade of resit.

Literature

Mandatory literature:
Fourier Analysis, an Introduction, by Elias M. Stein and Rami Shakarchi. Princeton Lectures in Analysis I. Princeton University Press, 2003, ISBN-13: 978-0691113845.

Additional Information Target Audience

Bachelor Mathematics Year 2

Recommended background knowledge

First year courses Single variable calculus, Multivariable calculus, Linear algebra, and (most importantly) Mathematical analysis.

From Molecule to Mind

Course Code	AM_1275
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	D.A. Maas
Examiner	prof. dr. H.D. Mansvelder
Teaching Staff	dr. G.J. Schenk, prof. dr. M. Verhage, prof. dr. H.D. Mansvelder, D.A. Maas, prof. dr. A.B. Smit
Teaching method(s)	Practical, Written partial exam, Lecture

Course Objective

Providing master students, independent of primary training, with a solid basis in molecular and cell biology, neurophysiology and functional neuroanatomy.

Course Content

The course will be used to brush up your knowledge of neurophysiology, molecular and cellular neurobiology and neuroanatomy. Lectures by established researchers on exciting and cutting-edge topics as well as a keynote symposium related to the different research programs on campus will be held to give you an impression of the research we do and to prepare you for your search for an internship project (which will start in February 2024).

Additional Information Teaching Methods

Lectures, study groups, assignments, practical sessions*, presentations, peer feedback, demonstrations.

*Please be aware the number of spots for the practicals is limited. In case the maximum number of students is exceeded, the course coordinator will decide who will be registered for the practicals, for the others there is an online practical available.

Method of Assessment

You will have to prepare presentations based on neuroanatomy related assignments and are expected to participate actively during practical sessions. You will receive a pass/fail grade for neuroanatomy. A 'pass' will be rewarded by a 30% correct starting grade for the final exam; an 'incomplete' will be rewarded by 0% (which means your maximum final grade for the course will be 70%). The final exam will consist of a digital multiple choice examination with two parts: cell biology and neurophysiology.

Entry Requirements

All students following the VU Master Neurosciences program are required to take this course.

Literature

- Alberts et al., Essential Cell Biology, 4th edition (2013; ISBN 9780815344544). Purves et al., Neuroscience, 5th edition (2012; ISBN 9780878936953). NB. More recent print editions of these books may also be used.

Additional Information Target Audience

First year VU Master Neurosciences

Recommended background knowledge

We strongly advise you to study these chapters before the course starts.NB. Alternatively, study equivalent chapters from more recent printings of these books:

- Alberts et al., Essential Cell Biology, 4th edition (2013; ISBN 9780815344544). Chapters 4, 5, 7, 8, 15 - 17
- Purves et al., Neuroscience, 5th edition (2012; ISBN 9780878936953). Chapters: 3 - 8.

From Protein to Cell

Course Code	AB_1052
Credits	6
Period	P2
Course Level	300
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Bald
Examiner	dr. D. Bald
Teaching Staff	dr. D. Bald
Teaching method(s)	Practical, Lecture

Course Objective

The student:

1. before starting an experiment prepares for the experiment and has an overview of the already existing

knowledge.
2. can correlate an open question with a suitable approach and can design an experiment.
3. can use insight into fundamental protein properties to accurately handle proteins and to carry out experimentation on proteins.
4. can link obtained results from experiments to the open question and to the already existing knowledge.
5. can present results from experiments in a report in the style of a brief scientific publication.

Course Content

The course consists of a mixture of lectures, practicals (in groups), and individual study. We will cover concepts/methods/techniques that you can use to study a broad range of relevant questions, e.g: how to study a protein and how to design an experimental strategy. In the laboratory practicals selected features of key proteins will be studied. The design of experiments to address a certain open question and understanding of experimental protocols are an important aspects of this course.

Additional Information Teaching Methods

Lectures: 8 h, practicals: 20 h, individual study: 140 h.

Method of Assessment

Written reports on the performed experiments (80%), assignment on CRIPSR-Cas (20%).

Literature

Lecture slides and experimental protocols. Any biochemistry textbook can be used for repetition.

Additional Information Target Audience

Students in the Minor Biomolecular Sciences.

Additional Information

This course is scheduled in the second half (weeks 5-8) of period 2.
A lab coat is mandatory for the lab practical lab work.
As this course mainly consists of practical exercises the examination is offered only once per study year.

The usage of generative AI in this course is not allowed.

Recommended background knowledge

Participation in course Experimental Biomolecular Sciences.

Functional Analysis

Course Code	XMM_400328
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Fundamentals of Bioinformatics

Course Code	X_405052
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	dr. H. Mouhib, dr. J.M. Urtasun Elizari, dr. ir. K.A. Feenstra, dr. D. Molenaar
Teaching method(s)	Seminar, Computer lab, Written partial exam, Lecture

Course Objective

Are you interested in Bioinformatics? Do you want to find out how biology can make an exciting application domain? Or do you want to learn what more you could do with your data, and with less effort? Enter here to start!

Fundamentals of Bioinformatics (FoB) is the starting course of the Bioinformatics master. It aims to give you a broad overview of important topics relevant to the field, with a focus on current open problems. You will be made aware of these open problems during practical sessions that will cover active research in a major Bioinformatics problem (mutation impact prediction). Based on your background, you will be assigned to separate classes where you will be working to fill gaps in your background knowledge in programming and/or biology.

After completing the course successfully:

- You can explain the fundamental principles of cell biology and the genetic information flow
- You can write short code to solve simple problems and complete skeleton scripts for biological applications
- You can read algebra and calculus equations
- You can become aware of gaps in your own background knowledge.
- You can address gaps in your background knowledge (programming, mathematics and molecular biology) and skills
- You are aware of the major issues when developing and benchmarking new methodology in bioinformatics research
- You can present and communicate a joined group projects carried out with students of diverse backgrounds (Interdisciplinarity) in a respectful way
- You can apply and understand fundamental methods in bioinformatics, e.g. sequence alignment, impact prediction, and to understand and use fundamental concepts in bioinformatics, e.g. homology, evolutionary conservation, and gene function

Fundamental knowledge:

Students will have the opportunity to level up their knowledge of molecular biology, maths (linear algebra and calculus) and programming (Python), according to their needs.

Course Content

Topics:

- Evolution, genomics, sequence alignment, Blast/PSI-Blast, FAIR data, benchmarking, machine learning, genome assembly, mutation impact prediction, protein stucture analysis, substitution matrices, scientific writing

Project:

- Group project to benchmark different mutation impact prediction approaches. The groups will be composed to include different BSc backgrounds.

Background knowledge (1 out of 3, see below):

- Molecular cell biology
- Programming and computational thinking in Python
- Calculus and linear algebra

Additional Information Teaching Methods

- 8 lectures (two-hour lecture in the morning, two days per week) Exercises during/in between lectures
- 12 computer practicals for the group project - compulsory attendance (online or on campus) (two hour sessions following the morning lectures, two days per week), supervised.
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Conversion classes for background knowledge:

- Molecular cell biology (7 x double lectures)
- Programming and computational thinking in Python (10 computer practicals)

Depending on your background, you will be assigned 1 of the 2 conversion classes. Students that are already well versed in both topics (molecular biology and programming) may be assigned to the following class, which is organised as part of the Introduction to Systems Biology:

- Calculus and linear algebra (10 maths classes)

Method of Assessment

- Project work [~40%] :
- Project report and/or presentation
- Digital exam to assess [~60%]:
- Lecture topics
- Concept learned during the project
- Concepts from conversion classes

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Further assessment and grading details will be posted on Canvas (resits and compensation rules)

Entry Requirements

Bachelor in any science discipline (including medicine) or strong programming background, and an interest in biological problems.

Literature

- course material (slides, scientific papers) on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics Garland Science 2008 ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will help you to set this up.

Recommended background knowledge

Programming in Python, Molecular Biology, some basics of calculus and linear algebra

Graph Symmetries and Combinatorial Designs

Course Code	XMM_0044
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

High-dimensional Probability

Course Code	XMM_0079
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Homotopy Type Theory

Course Code	XMM_0061
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath, mSFM

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Inclusieve Digitale Didactiek

Course Code	O_MIDD
Credits	3
Period	P2, P5
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S.M. Calor
Examiner	dr. S.M. Calor
Teaching Staff	dr. S.M. Calor
Teaching method(s)	Seminar, Lecture

Course Objective

De doordachte afstemming van complementaire face-to-face en online activiteiten moet duidelijk tot uiting komen in het ontwerp dat studenten leveren voor hun (nieuwe of bestaande cursus). Hiervoor gebruiken zij de VU Amsterdam Blended Learning (her) ontwerpsjabloon. Een belangrijk onderdeel hiervan betreft de motivatie die studenten gebruiken. Daarnaast lichten ze hun keuze voor online en face-to-face activiteiten toe en de volgorde hiervan, waarbij ze ook aandacht besteden aan constructive alignment; leerdoelen, leeractiviteiten en (formatieve en summatieve) testactiviteiten zijn op elkaar afgestemd. Inclusiviteit is een van de kernwaarden van de VU. Studenten gebruiken de VU Amsterdam Inclusiviteitskijkkader om de mate van inclusiviteit van het ontwerp in kaart te brengen. Ten slotte heeft de introductie van Generatieve Artificial Intelligence (AI) een enorm potentieel voor het onderwijs. Studenten gebruiken een VU Amsterdam doordachte didactiek bij de integratie van Generatieve AI in hun ontwerp.

Course Content

De coronaperiode heeft de inzet van digitale onderwijsvormen zoals online, blended en hybride onderwijs in een stroomversnelling gebracht. Naar aanleiding van het rapport Ruimte voor Onderwijs tijdens Corona en de motie Kwint is de wens ontstaan om meer inzicht te krijgen in de manier waarop digitale onderwijsvormen bijdragen aan het inhalen van coronavertragingen en het verbeteren van het welzijn van studenten. Daarnaast heeft de introductie van generatieve kunstmatige intelligentie (AI) een enorm potentieel voor het onderwijs. In het keuzevak “Inclusieve Digitale Didactiek” wordt aandacht besteed aan blended learning, inclusieve digitale didactiek en de integratie van generatieve AI in het onderwijs. Recente onderzoeksresultaten uit VU onderzoek worden in het keuzevak verwerkt.

De studenten van vandaag zijn opgegroeid in een wereld waar internet altijd aanwezig was, evenals mobiele telefoons en instant messaging. Ondersteunende leertechnologieën bieden een nieuwe manier om universitair onderwijs te ontwerpen, ontwikkelen en verzorgen en vormen een aanvulling op traditionele onderwijs- en leermethoden (Orton-Johnson, 2009). Als gevolg hiervan passen universiteiten en hogescholen hun financiering aan deze veelbelovende en nieuwe technologieën aan. De mogelijkheden voor communicatie, interactie en samenwerking worden uitgebreid, waardoor het hoger onderwijs naar het kantelpunt van een belangrijke transformatie wordt geduwd, waarvoor nieuwe vaardigheden van zowel studenten als docenten nodig zijn. Leertechnologieën kunnen ongekende onderwijsmogelijkheden bieden. Het vereist echter vakmanschap om effectieve blended learning, inclusieve lessen waarin generatieve AI in is geïntegreerd, te ontwerpen en ontwikkelen. Leraren moeten kennis hebben van inhoud, didactiek en technologie, evenals het vermogen om deze kennis zodanig toe te passen dat technologie het leren verbetert.

Additional Information Teaching Methods

Dit vak is opgebouwd uit een aantal colleges. Tussentijds bestuderen studenten relevante wetenschappelijke literatuur en werken zij aan hun onderwijs ontwerp voor een vak dat zij zelf geven. Tussendoor is er gelegenheid gebruik te maken van individuele coaching. Studenten presenteren hun tussentijds ontwerp aan elkaar en ontvangen van iedereen feedback. Deze peer feedback kan gebruikt worden ter voltooiing van het onderwijsontwerp. Tijdens de presentatie besteden studenten niet alleen aandacht aan hun ontwerp, zij lichten tevens hun analyse toe alsmede hoe het blended onderwijs, inclusief onderwijs waarin generatieve AI is geïntegreerd, te ontwikkelen, te implementeren en te evalueren. Zowel presentatie als ontwerp worden beoordeeld, resulterend in één eindcijfer.

Method of Assessment

De cijfers van studenten voor deze cursus worden berekend op basis van het blended learning ontwerp en het inclusief ontwerp met daarin generatieve AI geïntegreerd en hun presentatie hiervan.

Entry Requirements

Didactiek 1, Praktijk 1 (of vergelijkbaar)

Literature

Wordt bekend gemaakt via Canvas.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Custom Course Registration

Alleen studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Infinity-categories

Course Code	XMM_0062
Credits	8
Period	P1+2+3
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath, mSFM

Additional Information

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Interacting Particle Systems in the Lattice and Continuum

Course Code	XMM_0080

Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

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Introduction to Master Mathematics

Course Code	XM_0095
Credits	6
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. T.O. Rot
Examiner	dr. T.O. Rot
Teaching Staff	prof. dr. J. Hulshof, dr. T.O. Rot
Teaching method(s)	Lecture

Course Objective

At the end of this course the students will:

- be aware of the different research areas within mathematics and their mutual interactions;
- be aware of different job prospects with a degree in mathematics
- be able to make informed decision about their education, in particular with regard to the choice of a topic for their Master project, in a way that suits their career plans;
- have acquired appropriate reading and writing skills, and use these skills to evaluate information from various literature sources and produce scientific texts that are clear, consistent, and complete;
- be familiar with the basics of scientific research ethics and know at which stages in the preparation of their Master thesis they need to be paying particular attention to ethical issues (acknowledging other people's

work, openly sharing data and ideas…)

The course also specifically aims at enhancing the social cohesion and exchange of ideas between Master students following different study paths.

Course Content

The course will be centred around the following core elements:

- Introduction to mathematical research areas: problems, applications, and interactions
- Academic Reading and Writing
- Scientific Ethics
- Orientation on the career market

Additional Information Teaching Methods

The course will consist of meetings throughout the academic year. These meetings will be very different in nature: just to name a few examples, a meeting might consist of a scientific writing training, or it might require students to attend a research seminar and write a report about it. The exact schedule and content of the meetings will be made known in advance via Canvas. Attendance is compulsory and active participation is expected. Most of the course load is in the fall semester.

Method of Assessment

This course is assessed with a pass/fail.

Literature

Literature will be made available online.

Additional Information Target Audience

This course is obligatory for first year master students in Mathematics
(all study paths in Applied and Pure Mathematics).

Introduction to Systems Biology

Course Code	X_428565
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G. Stolf Jeuken
Examiner	dr. D. Molenaar
Teaching Staff	dr. M.T.M. Wortel, dr. D. Molenaar, prof. dr. F.J. Bruggeman, dr. G. Stolf Jeuken
Teaching method(s)	Seminar, Study Group, Written partial exam, Lecture

Course Objective

Introduction to Systems Biology is the starting course of the Bioinformatics and Systems Biology master (together with Fundamentals of Bioinformatics). To emphasise the importance of systems thinking biology, this course introduces the principles of systems biology, with a primary focus on the analysis of metabolic networks using stoichiometric modelling.

The course explores how genomic information can be translated into computational models to understand and predict cellular behaviour, linking genotype and phenotype. The course will see the application of these models in biotechnology and medicine, utilizing mathematical concepts (particularly linear algebra) and computational tools.

After completing the course successfully [Bloom taxonomy]:

- You understand the challenge of linking genomic information (genotype) to cellular behaviour (phenotype). [understand]
- You can write short code to solve simple problems and complete skeleton scripts for computational modelling of biological processes [apply]

- You can solve mathematical problems using algebra. [apply]
- You are able to represent biological information, specifically metabolic networks, in a computable format using mathematical concepts. [evaluate]
- You are able to use computational tools and algorithms to analyse metabolic networks. [apply]
- You appreciate the importance of experimental data and design in systems biology. [analyse]

This will enable you to understand and perform computational analysis related to:

- Systems Biology Research: Using metabolic network models to understand cellular growth under different conditions.
- Biotechnological Applications: Identifying genes to modify in microbial genomes for the production of valuable products like biofuels.
- Medical Applications: Identifying potential drug targets within the metabolism of pathogens or cancer cells for therapeutic intervention.

Fundamental knowledge:

Students will have the opportunity to level up their knowledge of molecular biology, maths (linear algebra and calculus) and programming (Python), according to their needs.

Course Content

The course will cover the following key topics:

- Genome Annotation and Network Reconstruction: Decoding genomic information to build metabolic networks.
- Stoichiometric Principles: Understanding chemical element and charge conservation in biochemical reactions, stoichiometric coefficients, and balancing reactions.
- Reaction Networks and Stoichiometric Matrix: Representing networks mathematically using the stoichiometric matrix and formulating mass balance equations.
- Steady-State Analysis: Concepts of steady-state, null space of the stoichiometric matrix, steady-state flux distributions, and the rank theorem.
- Linear Programming and Flux Balance Analysis (FBA): Introduction to FBA as a core computational method for analysing metabolism.

Conversion classes:

Depending on their needs, students can choose to attend conversion classes where they will be working to fill gaps in their knowledge in mathematics (linear algebra and differential equations), programming or biology.

Additional Information Teaching Methods

- Lectures
- Workgroups
- Conversion classes in biology or mathematics or programming

Method of Assessment

A written exam. To pass the exam, the grade must be 5.5 or higher. Retakes of the exam take place once a year. The last obtained pass grade will count as the final grade for the course.

Entry Requirements

Bachelor in any science discipline (including medicine), with an interest in applying mathematical techniques in biology. When in doubt please contact the coordinator.

Literature

A course syllabus will be provided

Additional Information Target Audience

Students with a bachelor degree in Biology, Bioinformatics, Mathematics, Physics or similar background

Recommended background knowledge

Mathematics at high-school level

Inverse Problems in Imaging

Course Code	XMM_0047
Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

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Lie Algebras

Course Code	XMM_0037
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Lie Groups

Course Code	XMM_400350
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/ Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Machine Learning Theory

Course Code	XMM_0002
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai

Examiner	
Teaching Staff	
Teaching method(s)	Written partial exam, Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Master Project Mathematics

Course Code	X_400355
Credits	36
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. J. Hulshof
Examiner	prof. dr. J. Hulshof
Teaching Staff	
Teaching method(s)	

Course Objective

The objectives of the master project are:

- to explore a mathematical research problem, or in case of an external project, to distil a mathematical problem formulation from the context of the host organisation,
- to study relevant papers from the (mathematical) literature, to combine those, and to add an original contribution,
- to put the results and conclusions in proper perspective, also in relation to results obtained by others,
- to present the research both in writing and in an oral presentation.

The student is expected to show:

- sufficient mathematical maturity,
- skills appropriate for a professional mathematician: technical mathematical skills, independence and initiative, originality and creativity and overview of the mathematical subject area.

Course Content

The Master's programme is concluded by an internal or external master project. An external project ("internship") is

carried out within a business, industry or research facility other than the departments of Mathematics.

For an internal research project, the student starts by identifying a research topic in consultation with his/her supervisor. This leads to a research plan, which is recorded in the 'master project planner' and a copy of this form is given to the master coordinator. The planner can be found on Canvas.

The project itself usually starts with a literature study, leads towards the boundaries of mathematical knowledge, and ideally culminates in original research by the student. The work is carried out by the student individually, while there are weekly or biweekly meetings with the supervisor to discuss progress and scientific questions. The work is presented in a master thesis and in a final (oral) presentation. (mandatory).

Additional Information Teaching Methods

Individual work; Either the student performs individual research or the student is an intern at a host organization.

Method of Assessment

Assessment is based on attitude and execution (35%), the written master thesis (50%) and the final presentation (15%)
The form used for the assessment of a master project can be found on Canvas.

Entry Requirements

A shortfall of 8EC is permissible, ie 76 EC of the master programme need to be completed before starting the final project.

Literature

Assigned individually.

Additional Information Target Audience

mMath

Additional Information

Please register for this course in VUnet, you then also have access to the information on Canvas.

If you are planning to do an external project, please contact the MSc Mathematics coordinator at least four months prior to the start.

Master Project Mathematics (T,E track)

Course Code	X_405037
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. J. Hulshof
Examiner	prof. dr. J. Hulshof
Teaching Staff	
Teaching method(s)	

Course Objective

The objectives of the master project are:

- to explore a mathematical research problem, or in case of an external project, to distill a mathematical problem formulation from the context of the host organization,
- to study relevant papers from the (mathematical) literature, to combine those, and to add an original contribution,
- to put the results and conclusions in proper perspective, also in relation to results obtained by others,
- to present the research both in writing and in an oral presentation.

The student is expected to show:

- sufficient mathematical maturity,
- skills appropriate for a professional mathematician: technical mathematical skills, independence and initiative, originality and creativity and overview of the mathematical subject area.

Course Content

The Master's program is concluded by an internal or external master project. An external project ("internship") is carried out within a business, industry or research facility other than the departments of Mathematics.

For an internal research project, the student starts by identifying a research topic in consultation with his/her supervisor. This leads to a research plan, which is recorded in the 'master project planner' and a copy of this form is given to the master coordinator. The planner can be found on Canvas.

The project itself usually starts with a literature study, leads towards the boundaries of mathematical knowledge, and ideally culminates in original research by the student. The work is carried out by the student individually, while there are weekly or biweekly meetings with the supervisor to discuss progress and scientific questions. The work is presented in a master thesis and in a final (oral) presentation. (mandatory).

Additional Information Teaching Methods

Either the student performs individual research or the student is an intern at a host organization.

Method of Assessment

Assessment is based on attitude and execution (35%), the written master thesis (50%) and the final presentation (15%). The form used for the assessment of a research project can be found on Canvas.

Entry Requirements

Before starting the project, the student in the Education track must have earned all other mathematics and elective credits of the programme except a shortfall of 8EC is permissible.

The student in the Teachers track must have earned all other mathematics credits of the programme except a shortfall of 8EC is permissible.

Literature

assigned individually

Additional Information Target Audience

mMath

Additional Information

Please register for this course in VUnet, you then also have access to the information on Canvas.

If you are planning to do an external project, please contact the MSc Mathematics coordinator at least four months prior to the start.

Mathematical Biology

Course Code	XMM_400504
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

After completion of the course, the student is able to

- read and understand the research literature about (deterministic) models of biological phenomena
- participate actively in projects that aim to model biological phenomena
- derive mathematical equations from bookkeeping considerations
- interpret mathematical results in the biological context that motivated the analysis; more precisely the point is that mathematical statements are translated into a relation between phenomena and the underlying mechanisms
- use formal arguments (based on differences in the time- or spatial scale of various mechanisms) to simplify equations in a meaningful way
- apply various analytical techniques to study phase portraits of planar ODEs representing ecological systems
- derive and analyse linear diffusion equations and their solutions
- apply bifurcation theory to study systems of nonlinear reaction-diffusion equations

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

- lectures (notes are in preparation and should be ready by the time the course is given) which explain and illustrate the methods while referring to other sources for detailed accounts of the underlying mathematical theory
- assignments which provide training in modelling and in the use of the methods. Students work on assignments, using pen and paper.

Method of Assessment

Grades are to a large extent based on the handed in written assignments and on oral presentations. Grading is based on 5 homework assignments and the final project. The average grade of the 5 home assignments will contribute 40% to the final grade. The written work on the paper will contribute another 40% and the remaining 20% will come from the oral presentation. Given the nature of this course, there is no resit exam.

Literature

Lecture notes will be provided by the instructors. See also the course website for the latest details: http://www.few.vu.nl/~rplanque/Onderwijs/MathBio/

Additional Information Target Audience

Mmath

Additional Information

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl. Registration required via https://elo.mastermath.nl/login.

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Recommended background knowledge

Basic knowledge about linear algebra, analysis, ODE, stochastic processes. (The key point, however, is the attitude: students should be willing to quickly fill in gaps in background knowledge.)

Mathematical Optimization

Course Code	XM_0051
Credits	6

Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R. Paradiso
Examiner	dr. R. Paradiso
Teaching Staff	dr. R. Paradiso, dr. A. Zocca
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

After completing this course:

- You can formulate a problem as an optimization model in a correct and efficient way.
- You understand when you need to use discrete decision variables and the impact of those in the effort required to solve problems.
- You can prove that a given model is convex, and you can characterize the optimal solutions and formulate the equivalent dual model.
- You can reformulate most of the convex models into Second Order Conic Optimization models that can be efficiently solved.
- You are able to formulate and solve optimization problems that do account for uncertainty, both with a stochastic and a robust approach. You should be able to translate (when possible) robust or stochastic requirements into equivalent but finite constraints that you can solve.
- You can solve small problems of all the types studied on paper and problems of practical size on the computer using the Python programming language.

Course Content

Mathematical optimization is among the most essential instruments of prescriptive analytics and is used to make optimal decisions based on quantitative arguments.

This course will make you familiar with translating practical problems into optimization models and solving those models. Despite the goal of modelling and solving such problems, this course teaches the fundamental results from the mathematics of optimization, including optimality conditions, duality, stochastic optimization, and robust optimization.

The course covers linear optimization and its generalizations (conic and convex optimization). We will also consider optimization under uncertainty. Optimization models will be implemented and solved in Python using both open-source solvers and commercial solvers that are free for education.

Additional Information Teaching Methods

The course consists of the following contact moments:

Lectures in 2 blocks of 1:45 per week.
Tutorials in 1 block of 1:45 per week.
Computer lab in 1 block of 1:45 per week.

Presence and active participation is expected during all the above.

Method of Assessment

Exam (60%)
Practical assignments (40%)
The practical assignments cannot be repeated. To pass:

- The grade for the exam needs to be 5.0 or higher,
- The weighted average needs to be 5.5 or higher.

Literature

Lecture notes, slides, and additional material published via Canvas

Additional Information Target Audience

Students of Business Analytics, Econometrics and Operations Research, or other master programs with a strong

quantitative accent interested in Mathematical Optimization theory and practice.

This course is not intended for students who followed the course Mathematical Optimization in their minor due to considerable overlap.

Explanation Canvas

The slides, assignments and additional information will be made available via Canvas.

Recommended background knowledge

A first course in Operations Research and Linear Algebra is required background.

Besides introduction to Operations Research and Linear Algebra you should be familiar with (basic) programming in Python.

Mathematical Statistics

Course Code	XB_0049
Credits	6
Period	P1+2
Course Level	300
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. F.H. van der Meulen
Examiner	prof. dr. ir. F.H. van der Meulen
Teaching Staff	prof. dr. ir. F.H. van der Meulen
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

Students become familiar with the concepts of a statistical model, estimation of parameters, testing statistical hypotheses and uncertainty quantification. This is considered both from the frequentist and Bayesian perspective. They learn about various methods for the construction of estimators, tests and confidence sets and about mathematical theory for the evaluation and comparison of such methods.

Course Content

Examples of typical statistical problems are the prediction of an important quantity on the basis of data, or the question whether there is a relation between certain variables. In mathematical statistics we view observed data as realisations of random variables. A statistical model is a collection of probability distributions that we interpret as possible distributions of the observations. This collection is typically parametrised by a finite-dimensional parameter which often has an interpretation that helps to imbue the data with meaning. Based on observations we aim to infer the unknown parameter. The two main paradigms to this are frequentist and Bayesian statistics. In both setups we discuss parameter estimation, uncertainty quantification and hypotheses testing. Finally, we develop mathematical theory that allows to assess and compare the performance of statistical procedures.

Additional Information Teaching Methods

Lectures and exercise sessions

Method of Assessment

Written midterm exam and written final exam. The final grade is either the average of both grades (both rounded to 2 decimals) or the grade for the final exam (whichever gives highest grade). The final exam is on all topics of the course.

The retake is a single exam on all topics of the course.

Literature

Compulsory:

- Statistical Theory -- a concise introduction-- 2nd edition, by F. Abramovich and Y. Ritov. CRC Press, 2023. This is the main text that will be used.

- Additional lecture notes.

Recommended additional reading:

- An introduction to mathematical statistics, by F. Bijma, M. Jonker, A. van der Vaart. Amsterdam University Press, 2017

Additional Information Target Audience

BSc Mathematics, Year 2

Recommended background knowledge

Necessary background: Calculus, Linear Algebra, Probability Theory.

Measure Theoretical Probability

Course Code	XMM_400244
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath, mSFM

Custom Course Registration

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Measure Theory

Course Code	X_401028
Credits	6
Period	P1+2
Course Level	300
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	dr. W. Kager
Examiner	dr. W. Kager
Teaching Staff	dr. W. Kager
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

After this course:

1. the student will know and understand the basic concepts of measure theory and the theory of Lebesgue integration.
2. the student will understand the main proof techniques in the field, and will also be able to apply the theory abstractly and concretely.
3. the student will be able to write elementary proofs himself, as well as more advanced proofs under guidance.
4. the student is able to use measure theory and integration in Riemann integration and calculus.
5. the student is able to work with Lebesgue measure and to exploit its special properties.

Course Content

We motivate and introduce the notion of a measure, that is, a way to assign a size to as many subsets as possible in an abstract space. It turns out that it is in general not possible to measure all sets, at least if one insists on countable additivity of the measure. This leads to the notion of a sigma-algebra. We show how one can obtain a unique measure on a sigma-algebra once certain basic properties are imposed.

Once we have defined measure, we can introduce and discuss so called measurable functions which, roughly speaking, form the class of functions which we will be able to integrate. We then introduce and study integration of these measurable functions with respect to a measure. We discuss (among other things) the monotone and dominated convergence theorems concerning the interchangeability of limit and integral, the substitution rule, and the relation of this new integral to the Riemann integral. We also discuss multi-dimensional Lebesgue measure, product measures, and the Tonelli-Fubini theorems.

The theory leads to a new perspective on integration of functions, which is not only more general than the Riemann setting when working on the real line, but also allows one to integrate in an abstract setting. This is of crucial importance for the development of functional analysis and probability theory.

Additional Information Teaching Methods

Lectures and tutorials.

Method of Assessment

Written final exam, and a written midterm exam after 7 weeks. The final exam will be 50% of the final grade, and the midterm exam will be 50% as well.

Literature

Rene L. Schilling: Measures, Integrals and Martingales, Cambridge University Press.

Additional Information Target Audience

3MAT, 3EOR

Recommended background knowledge

Single Variable Calculus, Multivariable Calculus and Mathematical Analysis (or equivalent).

Mechanics and Thermodynamics in the Cell

Course Code	X_422589
Credits	6
Period	P2
Course Level	400
Language Of Tuition	Dutch

Faculty	Faculty of Science
Course Coordinator	dr. D. Dulin
Examiner	dr. D. Dulin
Teaching Staff	dr. D. Dulin
Teaching method(s)	Seminar, Written partial exam

Course Objective

Below is provided a point-to-point list of the learning goals of the class that can be used as guidelines for the exams. At the end of this course, the students will:

- know the basic concepts of physics (in particular mechanics, statistical physics and thermodynamics), physical chemistry and mathematics to describe quantitatively biology on the cellular and molecular level.
- use these basic concepts to derive quantitative and predictive models that give a deeper insight into living systems.
- implement and compare different models to analyze quantitatively experimental data obtained from state-of-the-art microscopy approaches and formulate new biological models.

Course Content

1. Basic concepts in thermodynamics and statistical physics to describe biological process such as protein-ligand binding, and polymer conformation.
2. Basic concepts in mechanics to describe how biological beam responds to stress and strain. Combining statistical physics and mechanics to describe semi-flexible polymer such as DNA.
3. Basic concepts of hydrodynamics to understand how liquid flows in the channels, and how dimensions impact the nature of the flow and the motion of objects in liquid.
4. Derivation of the law of diffusion from a macroscopic and microscopic perspective and its application to describe small molecule motion, protein-ligand association kinetics, and molecular motors translocation.

Additional Information Teaching Methods

Lectures (4h per week) & Tutorials (2h per week)

Method of Assessment

Written exam (60% of final grade) + written tests (2x, 10% of final grade each) on part of the course contents + tutorial assessments (20% of final grade).

Literature

Phillips, R., Kondev, J., and Theriot, J., Physical Biology of the Cell. 2nd Edition New York: Garland Science, 2012 (ISBN 0815344503). Chapter 1 to 13 (except 9 and 11). (1st edition is also fine)

Additional Information Target Audience

3N, 3MNW mi-BB, 3WN, 3S

Additional Information

This course is part of the Minors Biomedische Beeldvorming and Basics of Bioinformatics and Systems Biology.
Mathematics: Calculus & Mathematische Methoden (or comparable)
Physics: basics of mechanics & thermodynamics

Recommended background knowledge

Mathematics: Calculus & Mathematische Methoden (or comparable)
Physics: basics of mechanics & thermodynamics

Metabolomics

Course Code	AB_1291
Credits	6
Period	P2
Course Level	300
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	prof. dr. P.E.G. Leonards
Examiner	prof. dr. P.E.G. Leonards
Teaching Staff	prof. dr. P.E.G. Leonards, dr. L.A. Baumann
Teaching method(s)	Seminar, Study Group, Practical, Written partial exam, Lecture

Course Objective

Molecular biology studies the functioning of cells and organisms at the molecular level, and is applied in many fields. Metabolomics is a rapidly emerging field in molecular biology, and is one of the so-called omics technologies (genomics, proteomics, metabolomics). Metabolomics measures small molecules – the metabolites – in the cell or organism with the aim of determining the health status. But it can also determine the influence of the environment on health, for example through diet or exposure to toxic substances. Examples of metabolites are amino acids, ATP, sugars, neurotransmitters, lipids, etc.

Metabolomics aims to measure the complete set of metabolites (the metabolome) in organisms and is widely used in the medical, toxicological, and biological field. Metabolites function in a network and are transported to the desired destination via the streets as in a city. Metabolites have a major influence on the phenotype, behavior but also on the development of organisms.

This course aims to provide insight into the principles of metabolism in organisms, how metabolomics can be used in practice, how metabolites can be determined in organisms, and how the large amount of data can be handled, processed, visualized and statistically analysed (Big data analysis). The course has a practical approach and consists of lectures followed by a metabolomics experiment (practical). The practical is an important part of the course in which the student will conduct an experiment with zebrafish eggs and determine metabolites in the zebrafish embryos. Another important part is the processing of the large amount of data generated by using data visualization and statistical tools (introduction to processing of Big Data).

The following topics are covered:

- What is metabolomics
- What are metabolic networks
- Basic principles and workflows for determining and processing metabolite data
- Methods for the analysis of metabolites
- Applications of metabolomics
- Setting up a metabolomics experiment
- Processing and analysis of large data (visualization, metabolite networks, statistical analysis)

The practical is an important part of the course to give the student a better insight into the various aspects of metabolomics. The student carries out a metabolomics experiment with zebrafish eggs, whereby the development from egg to embryo is monitored. The metabolites (lipids and neurotransmitters) are determined, and the data is processed, among other things by visualizing the data, creating metabolic networks and statistically processing the data.

The learning objectives at the end of the course are:

- The student has knowledge and insight into the topics covered from metabolomics in cells and organism: The interactions between metabolites, proteins and genes.
- The student has insight into the methods of measuring metabolites and the challenges of studying the metabolome.
- The student gains experience with chemical analysis methods and the analysis of large data sets.
- The student has practical experience in setting up and conducting a metabolomics experiment.
- The student can process and analyse collected experimental data and can do the interpretation.

Course Content

The course provides an introduction to metabolomics and will discuss and use the tools and techniques (practical) needed to conduct a metabolomics study. Various applications of metabolomics in biology, medical and toxicological fields will be discussed. A focus point in the course is the performance of a metabolomics experiment in which you will work with eggs of zebrafish. You follow the development from egg to embryo, determine lipids and neurotransmitters in the embryos, and process and analyse the large amount of data.

In this course a large number of fundamental and practical topics from metabolism and metabolomics are covered such as:

- Metabolic Networks
- Metabolomics (principles, workflows in metabolomics, applications)
- Introduction to chemical analysis methods (chromatography, mass spectrometry)

- Introduction to data processing (including visualization and Big data analysis)

In the practical, the students will apply the material from the lectures via a metabolomics experiment with zebrafish to understand the principles of metabolic networks, the use of analytical equipment to determine metabolites, and the processing and analysis of large data sets.

Additional Information Teaching Methods

Lectures (18 hours)
Metabolomics practical (28 hours)
Data processing (20 hours)
Presentation (16 hours)
Practicum report (8 hours)
Self-study (78 hours)

Method of Assessment

Examination (1/3 of the final grade, average grade at least 5.5).
Presentation (1/3 of the final grade, average grade at least 5.5).
Practicum report (1/3 of the final grade, average grade at least 5.5).

Attending practicals and working groups must be sufficient.

Literature

- Background literature, not needed, Metabolome analysis : an introduction. Silas G. Villas-Bôas, et al. 2007. ISBN-13: 978-0-471-74344-6. Chapters 1-5.

Additional Information Target Audience

Part of the minor Biomolecular Sciences.

Modern Cryptography

Course Code	XMM_0053
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/.

Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Modular Forms

Course Code	XMM_400599
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMATH

Custom Course Registration

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Multivariate Statistics

Course Code	XMM_0081
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Network Dynamics, Symmetry and Synchrony

Course Code	XM_0181
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Bick
Examiner	dr. C. Bick
Teaching Staff	dr. C. Bick
Teaching method(s)	Lecture

Course Objective

This MSc-level seminar has the following learning goals:

- Learn mathematics related to symmetry in dynamical systems, network dynamical systems, and synchrony.
- Read scientific texts at a research level.
- Create and organize lecture material.
- Give seminar talks to your fellow students that explain the mathematical concepts and ideas involved.
- Constructively participate in a research seminar by asking questions and giving feedback.

Course Content

Symmetries have important consequences for dynamical systems. A basic fact is that points that are fixed under the action of the symmetry are also invariant with respect to the flow. If the dynamical system describes the evolution of coupled nodes in a network, then symmetries of the network structure naturally translate into symmetries of the dynamical system. Moreover, examples of fixed point subspaces of permutation symmetries that permute nodes correspond to sets where nodes are synchronized.

In this seminar, we will start with an introduction to equivariant dynamical systems—dynamical systems with symmetry. To complement the basic theory, we will look at dynamical systems with network structure to elucidate how symmetry shapes synchronization properties of these systems with a view of the research literature.

Additional Information Teaching Methods

A seminar, for 2 hours per week during each week of periods 4 and 5. Each student gives one or two presentations.

If the number of participating students is too low to have a presentation every week, we might change some sessions into discussions of material that has to be read in advance.

Method of Assessment

The assessment will consist of three parts: (1) Preliminary lecture plan(s) to be discussed with the instructor ahead of the seminar presentation; (2) Seminar presentation(s); (3) Final extended lecture notes on the topic of the seminar. Weighting will be communicated through Canvas.

There will be a minimum attendance requirement in order to get a grade for this course. The exact requirement will be determined at the beginning of the seminar.

Due to the nature of the seminar, there will not be a resit.

Literature

The introduction to equivariant dynamical systems will include selected topics from:

- Golubitsky, M., & Stewart, I. (2002). The Symmetry Perspective (Vol. 200). Birkhäuser Verlag.

Research literature related to the more advanced topics will be made available through Canvas.

Further reading

- Golubitsky, M., Stewart, I., & Schaeffer, D. G. (1988). Singularities and Groups in Bifurcation Theory. Vol. II. Springer-Verlag.

Numerical Linear Algebra

Course Code	XMM_400329
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

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Numerical Methods

Course Code	X_401039
Credits	6
Period	P1+2
Course Level	300
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Avitabile
Examiner	dr. D. Avitabile
Teaching Staff	dr. D. Avitabile
Teaching method(s)	Seminar, Lecture

Course Objective

At the end of this course students will be able to:

- implement the main algorithms of numerical analysis correctly and efficiently in Matlab.
- perform numerical calculations to solve nonlinear algebraic problems, eigenvalue problems, interpolation problems, signal processing and ODEs.
- apply methods from numerical analysis in a scientific setting.
- evaluate the reliability of numerical methods.
- report comprehensively on the structure of her/his algorithms as well as the computations performed using her/his code.

Course Content

Numerical methods are used frequently in all areas of science, such as fluid dynamics, meteorology and financial risk management. Moreover, techniques from numerical analysis play an important role in mathematical research on differential equations, stochastics, optimization, etcetera. We focus on the main numerical methods from modern-day analysis and scientific computing.

The list of subjects includes:

- error analysis
- systems of nonlinear equations
- eigenvalue problems
- least square methods
- fast Fourier transform
- ordinary (and partial) differential equations (no prerequisites needed)

Applications include

- phone number recognition
- Google/Page rank algorithm
- data analysis
- curve following
- planet motions
- and more.

Additional Information Teaching Methods

Lectures (once a week, 1x2=2 hours) and computer labs (once a week, 1x2=2 hours). A number of Matlab assignments form an integral part of the course.

Method of Assessment

The final grade is based on a set of reports and computer codes that have to be handed in. The number and weights of the assignments will be specified on canvas at the start of the course.

Resit opportunities: if the weighted average of the submitted assignments fails to surpass the necessary 55% needed to pass the course, then there will be an option of re-submitting inadequate submissions. A re-submission cannot attain a score greater than 60%.

Literature

There is no mandatory literature.

The following book is recommended for the first half of the course: Scientific Computing with MATLAB and Octave by Alfio Quarteroni and Fausto Saleri, ISBN: 978-3642453663

Additional Information Target Audience

Bachelor Mathematics year 2

Recommended background knowledge

Necessary background: Linear algebra and Multivariable calculus

Numerical Methods for PDEs

Course Code	XMM_0072
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Numerical Methods for Stochastic Differential Equations

Course Code	XMM_0071
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	

Teaching method(s)

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Operator Algebras

Course Code	XMM_418062
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see
https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Optimal Transport

Course Code	XMM_0063
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See http://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See http://elo.mastermath.nl

Method of Assessment

See http://elo.mastermath.nl

Additional Information Target Audience

mMath, mSFM

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Optimization of Business Processes

Course Code	X_400422
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. G.M. Koole
Examiner	prof. dr. G.M. Koole
Teaching Staff	prof. dr. G.M. Koole
Teaching method(s)	Lecture

Course Objective

In this course students learn about applications of operations research in manufacturing and services, especially health care, call centers and revenue management.

After this course the students:

- are familiar with some areas in which business process optimization techniques are used;
- are familiar with the most common optimization techniques and tools and knows how to apply them;
- understand the business context and is familiar with the most common pitfalls.

Course Content

We deal with a number of important application areas of optimization:
production logistics/manufacturing, call centers, health care and revenue management.
For each area we present quantitative problems and discuss how they can be solved using mathematical models, mostly forecasting, deterministic optimization and stochastic processes. We also discuss a number of new models. Students solve representative problems from the field in homework assignments, which include all steps from data to decision. Several guest lectures are given by people from industry.

Additional Information Teaching Methods

Lectures and practical work.

Method of Assessment

Written examination, 3 individual assignments in R or python, and a book presentation.

The assignments and the presentation each count for 10%, the exam for 60%. The minimal grade for the exam is 5.0.

Literature

Lecture notes.

Additional Information Target Audience

mBA, mBA-D, mMath

Recommended background knowledge

Applied Stochastic Modeling or equivalent knowledge

p-Adic Numbers and Their Applications

Course Code	XM_0184
Credits	6
Period	P1+2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R.M.H. de Jeu
Examiner	prof. dr. R.M.H. de Jeu
Teaching Staff	prof. dr. R.M.H. de Jeu
Teaching method(s)	Lecture

Course Objective

The aim of this master seminar is to learn to read scientific texts at an advanced level, to create and organise material, to give seminar talks to your fellow students, and to be able to explain the mathematical concepts and ideas involved. It is also a goal to learn how to constructively participate in a seminar by asking questions and giving feedback. We focus on an advanced topic in algebra: the theory of p-adic numbers and p-adic analysis.

Learning goals:

- The student knows basic constructions and properties of p-adic fields (construction of Qp, finite extensions, ramification), some p-adic analysis (power series in one variable), and some applications (e.g., the Skolem–Mahler–Lech theorem on zeroes in solutions of linear recurrence relations).
- The student can independently learn part of themathematical material, and give a talk about it.
- The student can independently select an assignment on that part, provide model solutions, and mark the work handed in.

Course Content

We study p-adic fields, which have many applications in number theory and arithmetic algebraic geometry. We start with the construction and basic properties of the field of p-adic numbers, the completion of the rationals for a metric based on a (fixed) prime number p. We then develop the theory of its finite algebraic extensions, its algebraic closure, and the metric completion of the latter. We study power series with coefficients in such fields, which will include statements on their structure and number of zeroes, based on the sizes of their coefficients. We discuss some applications of p-adic numbers, which, depending on time and interest, may include solving Diophantine equations, determining zeros of recurrence sequences, and the rationality of the zeta function for hypersurfaces over finite fields.

Additional Information Teaching Methods

A seminar, for 2 hours per week during each week of periods 1 and 2. Each student gives one or two talks. The student also creates a corresponding assignment for the other students (which has to be approved by the coordinators), and grades the corresponding solutions of the other students based on (approved) detailed model solutions. This grading is checked at random by the coordinators.

Method of Assessment

The grade is determined as follows:

- 30% by the talk(s) of the student;
- 30% by the understanding of the material by the student (both for the talk and the model solution);
- 40% by the grades of the work handed in.

There will be a minimum attendance requirement in order to get a grade for this course. The exact requirement will be determined at the beginning of the seminar.

Due to the nature of the seminar, there will not be a resit.

Literature

There are several textbooks that can be used, for example

- A. Robert, A course in *p*-adic analysis
- N. Koblitz, *p*-adic number, *p*-adic analysis, and zeta-functions
- J. Neukirch, Algebraic number theory

Depending on the topic, we may work from one of those, or other sources.

Additional Information Target Audience

mMath

Additional Information

If the number of participating students is too low to have a presentation every week, we might change some sessions into discussions of material that has to be read in advance.

Recommended background knowledge

Basic linear algebra (vector spaces, subspaces, quotient spaces, dimension, linear maps, matrices), basic analysis in one variable (sequences, series and their convergence properties; continuous/differentiable functions, power series with ratio of convergence), basic topology (notion of topology, closure completion, metric) and some algebra (groups, rings, fields, quotient groups, quotient rings, Chinese remainder theorem for rings, field extensions, Galois theory for field extensions of finite degree, finite fields). There are many books that cover those; e.g., for the parts in field theory (where the prerequisites are most advanced) one can look Sections 13.1 through 14.4 of David S. Dummit & Richard M. Foote, "Abstract algebra" (3rd edition, John Wiley & Sons, 2003), as well as many other books.

Partial Differential Equations

Course Code	XMM_400330
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Pedagogische en Didactische verdieping Semester 1

Course Code	O_MPEDIVER_F
Credits	9
Period	P1+2+3
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	dr. B. de Vries, D.S. Hoefakker-Laanstra, prof. dr. M. Meeter, drs. A.J.C. Monquil, dr. I. Pauw, dr. H.B. Westbroek, dr. O.P.M. van Buuren, drs. M.M. Bout, drs. J.K.W. Riksen, drs. E.D. van Noort, drs. Y.G. Meindersma, drs. N.H. Ypenburg, N. el Abdellaoui MSc, drs. H.R. Goudsmit, dr. C.D.P. van Oeveren, dr. S.M. Calor, C.L. Geraedts, M. Beukering, drs. R. Cornelisse MSc, drs. L.J. van Well-van Grootheest, drs. M.A. Huijskes MA, prof. dr. ir. F.F.J. Hermans, dr. S. Donszelmann
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de verdiepende pedagogische en (vak)didactische kennis en

vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van doorlopende lessenreeksen, en voor het handelen als docent in complexe(re) situaties.

Course Content

De cursus Pedagogische en Didactische verdieping is onderdeel van de derde fase (fase III) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 3. Tijdens Pedagogische en Didactische verdieping worden de onderdelen algemene didactiek en pedagogiek (AD), peergroup (PG) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Pedagogische en Didactische verdieping is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Pedagogische en Didactische verdieping staan de volgende kernpraktijken centraal: (1) Leerlijnen ontwerpen en evalueren, (2) De puber leren kennen, (3) Rekening houden met gedrags- en leerkenmerken, (4) Je onderwijs systematisch evalueren, (5) Omgaan met diversiteit in je klas, (6) Differentiëren, (7) Verantwoordelijkheid geven, (8) Mentor zijn, en (9) Je professionele ruimte benutten. De cursussen Didactiek 1 en 2 vormen samen het basisdeel van de ULO; de cursus Pedagogische en Didactische verdieping kan gezien worden als het verdiepingsdeel waarbij we nader ingaan op (het begeleiden van) individuele leerprocessen en het verzorgen van passend onderwijs, en het leren van leerlingen in een groepsdynamische setting. In Pedagogische en Didactische verdieping komen meer complexe kernpraktijken en bijbehorende thema's aan de orde. Het (vak)didactisch denken en handelen strekt zich uit over langere perioden, en er is aandacht voor het ontwerpen van leerlijnen. Ook wordt de lio nadrukkelijker uitgedaagd een eigen visie op onderwijs te geven en uit te dragen. Zo is de lesmethode niet langer leidend, maar wordt van lio’s in toenemende mate verwacht zelf invulling te geven aan de inhoud en didactiek van de lessen (waarbij zowel bestaand als eigen materiaal kan worden gebruikt). Tenslotte zullen de (vak)didactische overwegingen die ten grondslag liggen aan de eigen visie onderbouwd moeten worden met relevante literatuur en eigen praktijkervaringen.

Het fysieke onderwijs van Pedagogische en Didactische verdieping concentreert zich op de maandag (de instituutsdag). Tijdens de (werk)colleges AD komen genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. Tot slot zijn er peergroupbijeenkomsten o.a. gericht op intervisie en het schrijven van de meesterproef.

Het onderdeel AD in de cursus Pedagogische en Didactische verdieping is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Pedagogische en Didactische verdieping wordt afgesloten met een (geschreven) meesterproef, bestaande uit een ontwerp voor een lessenreeks van vier tot zes lessen, waarbij rekening wordt gehouden met een specifieke casusleerling, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik. Een belangrijk onderdeel is ook de systematische evaluatie waarin één aspect van het ontwerp en het effect op het leerproces van de leerling onder de loep wordt genomen.

Entry Requirements

Voorwaardelijk voor afronding van Pedagogische en Didactische verdieping: een voldoende beoordeling (of vrijstelling) van Didactiek 2.

Literature

De volgende drie boeken zijn verplicht voor het onderdeel algemene didactiek en pedagogiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.
- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast is het volgende boek voor veel studenten een zinvolle bron:

- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Dit laatste boek is niet verplicht. Tijdens de eerste colleges zullen we kort iets over dit boek vertellen, en is er gelegenheid om het even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om het boek al dan niet aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Pedagogische en Didactische Verdieping Semester 2

Course Code	O_MPEDIVER
Credits	9
Period	P4+5+6
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. M.M. Bout, drs. R. Cornelisse MSc, C.L. Geraedts, dr. H.B. Westbroek, M. Beukering, dr. O.P.M. van Buuren, D.S. Hoefakker-Laanstra, dr. B. de Vries, dr. C.D.P. van Oeveren, drs. L.J. van Well-van Grootheest, drs. E.D. van Noort, dr. S.M. Calor, J. Quartel MA, N. el Abdellaoui MSc, prof. dr. M. Meeter, dr. I. Pauw, prof. dr. ir. F.F.J. Hermans, drs. Y.G. Meindersma, dr. S. Donszelmann, drs. A.J.C. Monquil, drs. N.H. Ypenburg, drs. J.K.W. Riksen, drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de verdiepende pedagogische en (vak)didactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van doorlopende lessenreeksen, en voor het handelen als docent in complexe(re) situaties.

Course Content

De cursus Pedagogische en Didactische verdieping is onderdeel van de derde fase (fase III) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 3. Tijdens Pedagogische en Didactische verdieping worden de onderdelen algemene didactiek en pedagogiek (AD), peergroup (PG) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Pedagogische en Didactische verdieping is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Pedagogische en Didactische verdieping staan de volgende kernpraktijken centraal: (1) Leerlijnen ontwerpen en evalueren, (2) De puber leren kennen, (3) Rekening houden met gedrags- en

leerkenmerken, (4) Je onderwijs systematisch evalueren, (5) Omgaan met diversiteit in je klas, (6) Differentiëren, (7) Verantwoordelijkheid geven, (8) Mentor zijn, en (9) Je professionele ruimte benutten. De cursussen Didactiek 1 en 2 vormen samen het basisdeel van de ULO; de cursus Pedagogische en Didactische verdieping kan gezien worden als het verdiepingsdeel waarbij we nader ingaan op (het begeleiden van) individuele leerprocessen en het verzorgen van passend onderwijs, en het leren van leerlingen in een groepsdynamische setting. In Pedagogische en Didactische verdieping komen meer complexe kernpraktijken en bijbehorende thema's aan de orde. Het (vak)didactisch denken en handelen strekt zich uit over langere perioden, en er is aandacht voor het ontwerpen van leerlijnen. Ook wordt de lio nadrukkelijker uitgedaagd een eigen visie op onderwijs te geven en uit te dragen. Zo is de lesmethode niet langer leidend, maar wordt van lio’s in toenemende mate verwacht zelf invulling te geven aan de inhoud en didactiek van de lessen (waarbij zowel bestaand als eigen materiaal kan worden gebruikt). Tenslotte zullen de (vak)didactische overwegingen die ten grondslag liggen aan de eigen visie onderbouwd moeten worden met relevante literatuur en eigen praktijkervaringen.

Het fysieke onderwijs van Pedagogische en Didactische verdieping concentreert zich op de maandag (de instituutsdag). Tijdens de (werk)colleges AD komen genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. Tot slot zijn er peergroupbijeenkomsten o.a. gericht op intervisie en het schrijven van de meesterproef.

Het onderdeel AD in de cursus Pedagogische en Didactische verdieping is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback..

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Pedagogische en Didactische verdieping wordt afgesloten met een (geschreven) meesterproef, bestaande uit een ontwerp voor een lessenreeks van vier tot zes lessen, waarbij rekening wordt gehouden met een specifieke casusleerling, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik. Een belangrijk onderdeel is ook de systematische evaluatie waarin één aspect van het ontwerp en het effect op het leerproces van de leerling onder de loep wordt genomen.

Entry Requirements

Voorwaardelijk voor afronding van Pedagogische en Didactische verdieping: een voldoende beoordeling (of vrijstelling) van Didactiek 2.

Literature

De volgende drie boeken zijn verplicht voor het onderdeel algemene didactiek en pedagogiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.
- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast is het volgende boek voor veel studenten een zinvolle bron:

- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Dit laatste boek is niet verplicht. Tijdens de eerste colleges zullen we kort iets over dit boek vertellen, en is er gelegenheid om het even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om het boek al dan niet aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Peergroup 1 Semester 1

Course Code	O_MLPEERGR_1
Credits	0
Period	P1+2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen – de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Course Content

In de peergroup staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Additional Information Teaching Methods

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de startproef en de basisproef, en komen terug in de praktijk.

Literature

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 1 onderdeel uitmaakt van de vakken Didactiek 1 en 2 dienen studenten apart voor dit onderdeel ingeschreven te staan.

Peergroup 1 Semester 2

Course Code	O_MLPEERG_1F
Credits	0
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	N. el Abdellaoui MSc, drs. Y.G. Meindersma, drs. H.R. Goudsmit
Teaching method(s)	Study Group

Course Objective

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen – de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Course Content

In de peergroup staat het handelen als 'professional' centraal. Studenten leren de regie te nemen over hun eigen

leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Additional Information Teaching Methods

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de startproef (Didactiek 1) en de basisproef (Didactiek 2), en komen terug in de praktijk.

Literature

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 1 onderdeel uitmaakt van de vakken Didactiek 1 en 2 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Peergroup 2 Semester 1

Course Code	O_MLPEERG_2F
Credits	0
Period	P1+2+3
Course Level	0
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	drs. H.R. Goudsmit, N. el Abdellaoui MSc, drs. Y.G. Meindersma
Teaching method(s)	Study Group

Course Objective

De student werkt in de sessies Peergroup (PG) aan het systematisch reflecteren op de eigen onderwijspraktijk en gebruikt hierbij onder meer kennis uit de school en opleiding, feedback van derden en beeldmateriaal van de eigen praktijkuitvoering. Tijdens de beenkomsten wordt ingegaan op belangrijke zaken t.a.v. de meesterproef, zoals de casusleerling en videoreflectie. De bijeenkomsten maken onderdeel uit van het vak Didactiek 3.

Course Content

In een aantal peergroupsessies op maandag (instituutsdag) staat het handelen als 'professional' centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

Additional Information Teaching Methods

Aantal werkgroepen op maandag (instituutsdag).

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de meesterproef (Didactiek 3).

Entry Requirements

Didactiek 1 en 2 (inclusief peergroup 1) afgerond (of vrijgesteld van).

Literature

Eventuele literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Peergroup 2 is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 2 onderdeel uitmaakt van het vak Didactiek 3 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Peergroup 2 Semester 2

Course Code	O_MLPEERGR_2
Credits	0
Period	P4+5+6
Course Level	0
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	dr. I. Pauw, drs. R. Cornelisse MSc, drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

De student werkt in de sessies Peergroup (PG) aan het systematisch reflecteren op de eigen onderwijspraktijk en gebruikt hierbij onder meer kennis uit de school en opleiding, feedback van derden en beeldmateriaal van de eigen praktijkuitvoering. Tijdens de beenkomsten wordt ingegaan op belangrijke zaken t.a.v. de meesterproef, zoals de casusleerling en videoreflectie. De bijeenkomsten maken onderdeel uit van het vak Didactiek 3.

Course Content

In een aantal peergroupsessies op maandag (instituutsdag) staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

Additional Information Teaching Methods

Werkgroepen.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de meesterproef (Didactiek 3).

Entry Requirements

Didactiek 1 en 2 (inclusief peergroup 1) afgerond (of vrijgesteld van).

Literature

Eventuele literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Peergroup 2 is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 2 onderdeel uitmaakt van het vak Didactiek 3 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Performance of Networked Systems

Course Code	X_405105
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R.D. van der Mei
Examiner	prof. dr. R.D. van der Mei
Teaching Staff	prof. dr. R.D. van der Mei
Teaching method(s)	Lecture

Course Objective

Students will acquire basic knowledge of:

- Performance issues in communication systems, a complex interplay between communication networks (TCP/IP, UDP), application servers (HTTP, QUIC), services and clients
- Performance models for wireless and mobile network protocols (WiFi, GSM, GPRS, UMTS, HSDPA, 4G, 5G, 6G and mobile ad-hoc networks)
- Performance engineering principles and management of traffic in IP-based networks
- Performance modeling of streaming and elastic traffic
- Quantitative models for planning capacity in communication networks

Students will gain experience in modelling, engineering and planning performance of networked systems, and will learn how to tackle performance problems arising in today's information and communication systems.

Course Content

Over the past few decades, information and communication technology (ICT) has become ubiquitous and globally interconnected. As a consequence, our ICT systems are expected to process huge amounts of digital information, which puts a tremendous burden on our ICT infrastructure. At the same time, our modern society has become largely dependent on the well-functioning of our ICT systems: large-scale system failures and perceivable Quality of Service (QoS) degradation may completely disrupt our daily lives and have huge impact on our economy. These developments raise the need for understanding the nature and source of performance problems, and for finding effective ways to make our networks robust and reliable.

Motivated by this, the course will address performance-related challenges and solution methods in communication networks. To this end, we focus on performance modeling, traffic management, QoS provisioning and capacity planning for both wired and wireless/mobile communication networks (such as WiFi, 2G, 3G, 4G, 5G, 6G networks, and mobile ad-hoc networks). We address questions like:

- How do wired and wireless communication network technologies work?
- How can we model their performance?
- How can we manage and control traffic flows and provision QoS in IP-based networks?
- How can we deal with the omnipresence of uncertainty in virtually all aspects of communication networks?
- What techniques are available to model, analyze and optimize performance?

Additional Information Teaching Methods

Classroom lectures and a practical take-home homework assignment.

Method of Assessment

The assessment will be based on the homework assignment (total 20%) and a written exam (80%). Homework assignments can be made in groups of one or two students.

Literature

- A textbook on "High Performance Browser Networking" by Ilya Grigorik, O-Reilly, 2013.
- A reader on Stochastic Performance Modelling.
- Individual papers on specific topics.

Additional Information Target Audience

mBA, mCS, mPDCS, mCLS, mEctrie

Explanation Canvas

The relevant course information will be shared via CANVAS.

Recommended background knowledge

The students should have basic knowledge of computer networks.

Practice 1

Course Code	O_MLPRAK_1F
Credits	6
Period	P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. M.A. Huijskes MA, prof. dr. ir. F.F.J. Hermans, D.S. Hoefakker-Laanstra, prof. dr. M. Meeter, N. el Abdellaoui MSc, drs. E.D. van Noort, M. Beukering, dr. H.B. Westbroek, C.L. Geraedts, dr. O.P.M. van Buuren, dr. S.M. Calor, drs. R. Cornelisse MSc, dr. S. Donszelmann, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, drs. Y.G. Meindersma, drs. M.M. Bout, dr. I. Pauw, drs. H.R. Goudsmit, drs. L.J. van Well-van Grootheest, drs. A.J.C. Monquil, dr. B. de Vries
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage voor Praktijk 1 omvat minimaal 48 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk gelden, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. Tijdens Praktijk 1 werkt de student o.a. toe naar het zelf geven van een hele les.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op

het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Additional Information Target Audience

Studenten met een baan (zij-instromers, onderwijstrainees etc) geven in dit stadium al zelfstandig les. Bij deze studenten ligt de nadruk bij de begeleiding vanuit de werkplekbegeleider op het niveau van didactisch handelen in de les.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 1

Course Code	O_MLPRAK_1
Credits	6
Period	P1
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	dr. C.D.P. van Oeveren, M. Beukering, drs. M.M. Bout, drs. A.J.C. Monquil, drs. N.H. Ypenburg, drs. H.R. Goudsmit, drs. J.K.W. Riksen, dr. H.B. Westbroek, drs. R. Cornelisse MSc, dr. S. Donszelmann, N. el Abdellaoui MSc, D.S. Hoefakker-Laanstra, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, dr. I. Pauw, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, dr. O.P.M. van Buuren, C.L. Geraedts, prof. dr. M. Meeter, drs. E.D. van Noort, dr. S.M. Calor, drs. M.A. Huijskes MA
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage voor Praktijk 1 omvat minimaal 48 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk gelden, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. Tijdens Praktijk 1 werkt de student o.a. toe naar het zelf geven van een hele les.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Additional Information Target Audience

Studenten met een baan (zij-instromers, onderwijstrainees etc) geven in dit stadium al zelfstandig les. Bij deze studenten ligt de nadruk bij de begeleiding vanuit de werkplekbegeleider op het niveau van didactisch handelen in de les.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 2

Course Code	O_MLPRAK_2F
Credits	9
Period	P5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	dr. I. Pauw, prof. dr. M. Meeter, dr. H.B. Westbroek, M. Beukering, dr. C.D.P. van Oeveren, C.L. Geraedts, D.S. Hoefakker-Laanstra, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, drs. M.M. Bout, drs. R. Cornelisse MSc, N. el Abdellaoui MSc, drs. H.R. Goudsmit, dr. B. de Vries, dr. S.M. Calor, drs. A.J.C. Monquil, drs. J.K.W. Riksen, drs. N.H. Ypenburg, prof. dr. ir. F.F.J. Hermans, drs. M.A. Huijskes MA, drs. E.D. van Noort, dr. O.P.M. van Buuren, dr. S. Donszelmann
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. 2 van de stage omvat minimaal 72 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. De student werkt o.a. toe naar het zelf geven van een kleine lessenserie.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de module geldt:

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’.

Voor de master geldt:

De werkplekbegeleider beoordeelt Praktijk 2 aan de hand van het Beoordelingsformulier Praktijk.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Additional Information

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 2

Course Code	O_MLPRAK_2
Credits	9
Period	P2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. A.J.C. Monquil, M. Beukering, drs. E.D. van Noort, dr. H.B. Westbroek, prof. dr. ir. F.F.J. Hermans, D.S. Hoefakker-Laanstra, C.L. Geraedts, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, dr. S. Donszelmann, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, drs. M.M. Bout, dr. B. de Vries, dr. I. Pauw, drs. H.R. Goudsmit, dr. S.M. Calor, dr. O.P.M. van Buuren, drs. R. Cornelisse MSc, N. el Abdellaoui MSc, prof. dr. M. Meeter, drs. M.A. Huijskes MA, drs. N.H. Ypenburg
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 2 van de stage omvat minimaal 72 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. De student werkt o.a. toe naar het zelf geven van een kleine lessenserie.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor een modulestudent geldt het volgende:

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn

oordeel op deze ‘examenles’.

Voor een masterstudent geldt het volgende:

Praktijk 2 wordt beoordeeld door de werkplekbegeleider.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 3

Course Code	O_MLPRAK_3F
Credits	15
Period	P1+2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. H.R. Goudsmit, dr. B. de Vries, D.S. Hoefakker-Laanstra, C.L. Geraedts, drs. R. Cornelisse MSc, drs. Y.G. Meindersma, R.B. van der Vos, drs. N.H. Ypenburg, M. Beukering, drs. A.J.C. Monquil, drs. E.D. van Noort, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, dr. O.P.M. van Buuren, dr. I. Pauw, dr. C.D.P. van Oeveren, R.M.C. Canisius, drs. M.M. Bout, drs. J.K.W. Riksen, prof. dr. ir. F.F.J. Hermans, prof. dr. M. Meeter, dr. H.B. Westbroek, dr. S. Donszelmann
Teaching method(s)	Seminar

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 3 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 3 van de stage omvat minimaal 120 klassencontacturen (van 50 minuten). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. De student gaat meer en meer zelf(standig) lesgeven. De voorbereiding en evaluatie worden samen met de werkplekbegeleider gedaan. Ten minste 60 van de 120 lesuren worden door de student zelf verzorgd.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, dienen minimaal 120 lesuren door de student zelf te worden verzorgd, waarvan 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo). Het moet in ieder geval mogelijk zijn om de lessenreeks die ten tijde van Didactiek 3 wordt ontworpen (dat parallel loopt aan Praktijk 3) uit te voeren in een bovenbouwklas. Binnen Praktijk 3 zal daarom de bovenbouw veelal centraal staan.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. Ook met onderwerpen die bij keuzevakken op het instituut ter sprake komen zal in de praktijk worden geëxperimenteerd. De werkplekbegeleider wordt in ieder geval per mail en via de wpb-website op

de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur ook met de wpb en neemt vragen, ervaringen en inzichten mee naar de instituutsdag.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 3 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’. De VU-mentor legt ook een lesbezoek af (dit kan dezelfde ‘examenles’ zijn, maar hoeft niet).

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 3. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 3 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 3: een voldoende beoordeling van Praktijk 2 en Didactiek 2.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 3

Course Code	O_MLPRAK_3
Credits	15
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. E.D. van Noort, drs. N.H. Ypenburg, prof. dr. ir. F.F.J. Hermans, drs. Y.G. Meindersma, dr. I. Pauw, dr. S. Donszelmann, D.S. Hoefakker-Laanstra, dr. S.M. Calor, dr. H.B. Westbroek, dr. O.P.M. van Buuren, M. Beukering, N. el Abdellaoui MSc, dr. B. de Vries, drs. M.M. Bout, drs. R. Cornelisse MSc, drs. A.J.C. Monquil, dr. C.D.P. van Oeveren, prof. dr. M. Meeter, drs. H.R. Goudsmit, drs. L.J. van Well-van Grootheest, C.L. Geraedts, drs. J.K.W. Riksen
Teaching method(s)	Lecture

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 3 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een

belangrijke rol. Praktijk 3 van de stage omvat minimaal 120 klassencontacturen (van 50 minuten). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. De student gaat meer en meer zelf(standig) lesgeven. De voorbereiding en evaluatie worden samen met de werkplekbegeleider gedaan. Ten minste 60 van de 120 lesuren worden door de student zelf verzorgd.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, dienen minimaal 120 lesuren door de student zelf te worden verzorgd, waarvan 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo). Het moet in ieder geval mogelijk zijn om de lessenreeks die ten tijde van Didactiek 3 wordt ontworpen (dat parallel loopt aan Praktijk 3) uit te voeren in een bovenbouwklas. Binnen Praktijk 3 zal daarom de bovenbouw veelal centraal staan.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt in ieder geval per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur ook met de wpb en neemt vragen, ervaringen en inzichten mee naar de instituutsdag.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 3 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’. De VU-mentor legt ook een lesbezoek af (dit kan dezelfde ‘examenles’ zijn, maar hoeft niet).

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 3. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 3 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 3: een voldoende beoordeling van Praktijk 2 en Didactiek 2.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Praktijkonderzoek

Course Code	O_MLPROZ
Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. H.B. Westbroek
Examiner	dr. H.B. Westbroek
Teaching Staff	dr. H.B. Westbroek
Teaching method(s)	Study Group, Lecture

Course Objective

Aan het eind van deze cursus heeft de student:

1. Kennis van en inzicht in het opzetten, uitvoeren en presenteren van kleinschalig actieonderzoek;
2. Vaardigheid in het in samenwerking met experts op het instituut en op school opzetten, uitvoeren en presenteren van kleinschalig actieonderzoek.

Concreet houdt dit in:

- Het vertalen van een praktijkprobleem naar een onderzoeksvraag;
- Het ontwerpen en uitvoeren van een kleinschalige dataverzameling;
- Het maken van een analyseplan en uitvoeren van (kwalitatieve en/of kwantitatieve) analyse;
- Het rapporteren van de gebruikte methode, de resultaten, conclusie en discussie in een presentatie.

Course Content

In het keuzevak Praktijkonderzoek staat het leren toepassen van onderzoeksvaardigheden in de onderwijspraktijk centraal. In dit keuzevak leer je een actieonderzoek opzetten en uitvoeren ten behoeve van je eigen professionalisering. Vaak wordt een professionaliseringsvraag breder gedeeld in je sectie/school. Actieonderzoek is een vorm van onderzoekend leren. Je gebruikt theorie en bevindingen uit onderwijsonderzoek om jouw praktijkprobleem te analyseren en om ontwerp/voornemens te formuleren. Je leert hoe je in je klassen data kan verzamelen over de invloed van jouw ontwerp, op de leerprocessen van je leerlingen.

Additional Information Teaching Methods

introductiecollege (1,5 uur), werkgroepen/intervisiesessies (7,5 uur), zelfwerkzaamheid en individuele begeleidingsgesprekken (157,5 uur), presentaties (1,5 uur).

Method of Assessment

Praktijkonderzoek wordt afgesloten met een presentatie. Tussentijds dient ook een onderzoeksplan te worden ingeleverd dat wordt beoordeeld (go/no go).

Entry Requirements

Vereiste voorkennis: Didactiek 2, Praktijk 1 en onderzoekservaring op bachelor/masterniveau in het eigen domeinvak.

Literature

- Literatuur uit Didactiek 1 en Didactiek 2 vormt een basis.
- Literatuur wordt aangeboden in de vorm van wetenschappelijke artikelen (CANVAS).
- Van studenten wordt verwacht dat ze aanvullend zelf literatuur zoeken en gebruiken om hun theoretisch kader en onderzoeksmethode te kunnen uitwerken en onderbouwen.

Additional Information Target Audience

Praktijkonderzoek is een van de specialisaties binnen de ULO lerarenopleiding.

Explanation Canvas

Alle informatie over de opzet van het vak, de bijeenkomsten, toetsing en literatuur wordt aangeboden via CANVAS. Communicatie gebeurt via CANVAS (discussieomgeving; mail).

Quantum Computing

Course Code	XMM_0013
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	

Teaching method(s)

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

-

Method of Assessment

-

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Quantum Information Theory

Course Code	XMM_0024
Credits	8
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Queueing Theory

Course Code	XMM_400397
Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Riemann Surfaces

Course Code	XMM_400325
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Scheduling

Course Code	XMM_400396
Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath, mBA

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/.

Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Selected Areas in Cryptology

Course Code	XMM_40011
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Semidefinite Optimization

Course Code	XMM_0015
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Sequential Monte Carlo Methods

Course Code	XM_0182
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. F.H. van der Meulen
Examiner	prof. dr. ir. F.H. van der Meulen
Teaching Staff	prof. dr. ir. F.H. van der Meulen
Teaching method(s)	Lecture

Course Objective

The aim of this master seminar is to learn how to read scientific texts at a research level, create and organise material, to give seminar talks to your fellow students, and to be able to explain the mathematical concepts and ideas involved. It is also a goal to learn how to constructively participate in a research seminar by asking questions and giving feedback.

This seminar aims to provide an introduction to the theory and practice of Sequential Monte Carlo (SMC) methods, a versatile class of algorithms for Bayesian inference in dynamic and latent variable models. Based on the first ten chapters of "An Introduction to Sequential Monte Carlo Methods" by Chopin and Papaspiliopoulos, the course provides a systematic foundation in the core principles of SMC, including importance sampling, resampling, degeneracy, and variance analysis, with a focus on both discrete- and continuous-time state-space models.

The course progresses from basic SMC algorithms to more advanced topics such as backward smoothing, adaptive resampling, and effective sample size diagnostics. Students will gain hands-on experience with implementing SMC algorithms and will critically examine their theoretical properties, computational efficiency, and practical limitations.

Course Content

Learning goals:

- Describe the structure of state-space models and formulate inference problems within them.
- Learn the structure of Feynman-Kac models.
- Implement basic Sequential Importance Sampling (SIS) and Sequential Importance Resampling (SIR) algorithms.
- Explain and quantify particle degeneracy and variance in Monte Carlo estimators.
- Apply smoothing techniques such as backward simulation and forward-filtering backward-sampling (FFBS).
- Use diagnostics such as effective sample size (ESS) to monitor and adapt SMC algorithms.
- Analyze the convergence properties and asymptotic behavior of SMC estimators.
- Design and evaluate adaptive resampling strategies to improve algorithmic performance.

- Apply SMC methods to simple real-world or simulated data problems.
- Understand the theoretical assumptions and limitations of SMC in high-dimensional settings.
- Prepare for more advanced topics such as Particle MCMC and SMC samplers.

Additional Information Teaching Methods

A seminar, for 2 hours per week during each week of periods 1 and 2. Each student gives one or two talks. At the start of the course there are lectures.

Method of Assessment

Depending on the number of students, the grade will consist of the average of the grades for presentation and an oral examination. There will be a minimum attendance requirement in order to get a grade for this course. The exact requirement will be determined at the beginning of the seminar. Due to the nature of the seminar, there will not be a resit.

Entry Requirements

Solid understanding of probability theory and statistics at bachelor level mathematics.

Literature

"An Introduction to Sequential Monte Carlo Methods"(Springer) by Chopin and Papaspiliopoulos

Additional Information Target Audience

If the number of participating students is too low to have a presentation every week, we might change some sessions into discussions of material that has to be read in advance.

Set Theory

Course Code	XMM_418035
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see
https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Statistical Data Analysis

Course Code	X_401029
Credits	6
Period	P4+5
Course Level	300
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. M.A. Corstanje
Examiner	dr. ir. M.A. Corstanje
Teaching Staff	dr. ir. M.A. Corstanje
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

After students have followed the course, they should be able to

- make use of QQ-plots, symplots, histograms, boxplots, goodness-of-fit tests, etc. to find a suitable model distribution for a dataset at hand;
- estimate unknown model parameters (e.g. location and/or scale) and underlying density function;
- decide which statistical method is preferred to draw conclusions about the population underlying the data, for example using hypothesis tests and confidence intervals, while taking characteristics of the dataset into account (e.g. knowing when to use a nonparametric test);
- apply tests for location parameters, stochastic order, or equality of distributions in two-sample problems, and be able to assess the asymptotic relative efficiency of tests;
- using R, apply resampling methods such as the bootstrap or random permutation to find characteristics of a statistic even if no model assumptions are made;
- analyze the relationship between two (or more) variables in a given dataset with the help of rank-based correlation tests, chi-square tests for contingency tables, or multiple linear regression. In the context of multiple linear regression, students should be able to select variables for the linear regression model and identify potentially influential observations;
- communicate with colleagues about statistical topics, e.g. by solving the assignments in groups of two students and by talking to other students, teaching assistants, and the teacher of the course;
- create informative and scientifically appropriate reports which include relevant text, graphics and (if necessary) code. Reports should use appropriate language and be concise yet complete, as well as tidy, self-explanatory and (if applicable) reproducible.

Course Content

This is an advanced level statistical data analysis course that builds on introductory courses on probability and statistics, e.g. Statistics (X_400004) and Probability Theory (X_400622). The course introduces the students to several widely used statistical models and methods, and the students learn how to apply these tools to real data with the use of the statistical software R. The following topics are covered:

- summarizing data;
- investigating the distribution of data;
- density estimation;
- bootstrap;
- nonparametric methods;
- two-sample problems;
- contingency tables;
- multiple linear regression.

The course is a combination of theory (lectures) and practice (exercise classes), with the aim to directly link the theory to the practice of statistical data analysis.

Additional Information Teaching Methods

Lectures (13x2h; once per week), computer classes (13x2h; once per week).

Attendance is not mandatory but strongly recommended.

Method of Assessment

- 40% of the final grade consists of an average of assignment grades, the other 60% consists of the final exam grade. Both of these components have to be at least 5.5 to pass, students who fail either component are allowed to write the resit exam;
- Assignments consists of both homework reports (to be solved in groups of at most two students) and Canvas quizzes (to be submitted individually). Wherever necessary, assignments should be solved with the help of the programming language R;
- There is no resit possibility for the assignments themselves (see next bullet point);
- If the resit exam is written, the final grade will be the best one between a) the resit grade itself and b) a weighted-average grade where 40% consists of an average of assignment grades and 60% consists of the resit exam grade.

Literature

Lecture notes (Syllabus), mandatory reading.

Additional Information Target Audience

Business Analytics Year 2, Mathematics Year 2 or 3

Additional Information

Language of tuition: English.

Recommended background knowledge

The required knowledge has been obtained if the students had previously passed the VU courses Statistics (X_400004) / Mathematical Statistics (XB_0049) and Probability Theory (X_400622), or equivalent courses.

Statistical Models

Course Code	X_400418
Credits	6
Period	P1+2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.N. Belitser
Examiner	dr. E.N. Belitser
Teaching Staff	dr. E.N. Belitser
Teaching method(s)	Seminar, Lecture

Course Objective

The goals of this course are to get acquainted with some of the most commonly used statistical models, to learn how to apply these models in valid settings, and to understand the basic theory behind these models.

Course Content

ANOVA models, nonlinear regression models, generalized linear models, time series (possibly some related topics).

Additional Information Teaching Methods

Lectures and tutorials.

Method of Assessment

Assignments and examination.

Literature

Lecture slides and lecture notes "Statistical Models" by M.C.M. de Gunst.

Additional Information Target Audience

MSc BA, MSc Mathematics

Additional Information

Students will use the statistical package R (www.r-project.org) for data analysis.

Explanation Canvas

All the information about the course (and group enrollment) will be available on Canvas.

Recommended background knowledge

Course(s) on statistics and probability (such as Statistics X 400004 and Probability Theory X_400622 or their equivalents) and a basic course on Linear Algebra are required; experience with statistical package R is strongly recommended.

Stochastic Integration

Course Code	XMM_0049
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

-

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Stochastic Processes

Course Code	XMM_400339
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

-

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

-

Method of Assessment

-

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Stochastic Processes for Finance

Course Code	X_400352
Credits	6
Period	P1+2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. P.J. de Andrade Serra
Examiner	dr. P.J. de Andrade Serra
Teaching Staff	dr. P.J. de Andrade Serra
Teaching method(s)	Seminar, Lecture

Course Objective

After this course, the student will:

- be acquainted with the notion of pricing derivatives and arbitrage, and with the basics of discrete stochastic processes, including the concepts of martingales; be able to price derivatives in the binomial model.
- be familiar with the basics of continuous time stochastic processes, concepts of measure theory, Brownian motion, martingales; be able to verify whether the process is a Brownian motion and/or is a martingale.
- be familiar with the notions of stochastic integral Ito and SDE, the Ito's formula; be able to compute stochastic integrals, to use the Ito formula, and to derive properties of stochastic integrals and the solutions of SDEs.

- be familiar with the Black-Sholes model; be able to price options by the no-arbitrage and Girsanov's theorem, to compute the hedging portfolio and the Greeks, and to make connections to the corresponding Black-Scholes PDEs.
- be familiar with the extended Black-Sholes models; be able to price derivatives in those models, to make connection with the corresponding PDEs, and to apply the Black-Scholes machinery to the interest rates models.

Course Content

Financial institutions trade in risk, and it is therefore essential to measure and control such risks. Financial instruments such as options, swaps, forwards, etc. play an important role in risk management, and to handle them one needs to be able to price them. This course gives an introduction to the mathematical tools and theory behind risk management.

A "stochastic process" is a collection of random variables, indexed by a set T. In financial applications the elements of T model time, and T is the set of natural numbers (discrete time), or an interval in the positive real line (continuous time). "Martingales" are processes whose increments over an interval in the future have zero expectation given knowledge of the past history of the process. They play an important role in financial calculus, because the price of an option (on a stock or an interest rate) can be expressed as an expectation under a so called martingale measure. In this course we develop this theory in discrete and continuous time. Most models for financial processes in continuous time are based on a special Gaussian process, called Brownian motion. We discuss some properties of this process and introduce "stochastic integrals" with Brownian motion as the integrator. Financial processes can next be modelled as solutions to "stochastic differential equations". After developing these mathematical tools we turn to finance by applying the concepts and results to the pricing of derivative instruments. Foremost, we develop the theory of no-arbitrage pricing of derivatives, which are basic tools for risk management.

Additional Information Teaching Methods

Lectures (2 x 45min) and Tutorials (2x45). About 13 meetings of each are planned.

Method of Assessment

Two assignments covering computation and theory (up to 30% of final grade) and written examination/resit examination (at least 70% of final grade). There is no resit possibility for the assignments.

Literature

Lecture notes contain all of the material.
Additional, optional literature:

- Shreve, "Stochastic Calculus for Finance I: The Binomial Asset Pricing Model", Springer;
- Shreve, "Stochastic Calculus for Finance II: Continuous-time models", Springer.

Additional Information Target Audience

mBA, mBA-D, mMath, master Econometrics.

Additional Information

A significant part of the course is used to introduce mathematical subjects and techniques like Brownian motion, stochastic integration and Ito calculus. In view of this, the course is NOT meant for students who already followed the master course "Stochastic Integration" or "Stochastic differential equations". On the other hand, after completing this course, students may be motivated to follow other courses (like the two mentioned above) where stochastic calculus is treated in a deeper and more rigorous way.

Recommended background knowledge

Probability Theory (e.g., X_400622), Mathematical Analysis (e.g., XB_0009), Measure Theory (e.g., X_401028 or Mastermath course Measure Theoretic Probability)

Structural Bioinformatics

Course Code	X_405019
Credits	6
Period	P4

Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	R. Kumar
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	R. Kumar, dr. ir. K.A. Feenstra, dr. J.M. Urtasun Elizari, dr. H. Mouhib
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

Why Structural Bioinformatics?

The function of a protein is determined by its three-dimensional structure, and therefore structural information is crucial for understanding the working of proteins. However, experiments, prediction and simulation of protein structures remain difficult. This course will provide you with an overview of existing computational techniques, to validate, simulate, predict and analyse protein structures. More importantly, it will provide practical knowledge about how and when to use such techniques.

After completing the course successfully,

- You understand the basics of statistical thermodynamics and the relation to protein structures, evolution and function [Knowledge and understanding]
- You are able to evaluate protein structures with knowledge of their experimental source, bias and validation techniques [Applying knowledge and understanding]
- You are able to compare different protein structures, and evaluate similarity [Applying knowledge and understanding]
- You know how and when to use structure prediction methods [Making judgements; Applying knowledge and understanding]
- You can perform calculations on protein structures to gain new biological insights [Applying knowledge and understanding]
- You are able to compare different simulation techniques for biological macro-molecules, and be able to analyse the simulated data computationally [Applying knowledge and understanding]
- You are able to read and understand scientific papers in the field of Structural Bioinformatics [Communication; Lifelong learning skills]
- You are able to independently search for answers to open problems and ask a question about something that is unclear [Communication, Lifelong learning skills]

Course Content

Theory:

- Protein and DNA structure sources
- Experimental methods
- Structure validation
- Protein fold prediction (from homology modelling to ab initio prediction)
- Structural classification and structural alignment
- Protein folding and energetics
- Molecular Dynamics & Monte Carlo simulation
- Function from structure

Practical:

- Obtaining geometric features from PDB files
- Homology modelling with Modeller
- Protein interaction as a 'computational experiment' (simulation)

Additional Information Teaching Methods

- 13 lectures (2 two-hour lectures per week)
- Exercises during/in between lectures
- 12 computer practicals (2 two-hour sessions per week)
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Method of Assessment

The final grade for this course will consist of 50% practical work and 50% theoretical assessment.

Practical Assignments: (50%)
(assessment of practical assignments will be adjusted based on ongoing developments in generative language models)

Theoretical: (50%)

- Oral, written or digital exam (depending on number of course students) to assess lecture topics.
- As part of the exam research papers on Structural Bioinformatics topics need to be analysed in detail.
- You will be prepared for your exam through exercises and paper discussions during the lectures, as well as through the practical assignments.

Assessment and grading details will be posted on Canvas.

Entry Requirements

Bachelor in any science discipline (including medicine), with an interest in molecular biology and applying algorithmic approaches to protein structures.
Experience with programming (preferably python).
Basic understanding of calculus and of organic chemistry.
When in doubt please contact the coordinator.

Literature

- course material on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum. Understanding Bioinformatics. Garland Science 2008 ISBN-10: 0-8153-4024-9
- book "Introduction to Structural Bioinformatics": The intro is published on ArXiv [https://arxiv.org/abs/1801.09442], and the chapters are available from the table of contents. The book will also be posted on canvas.

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Artificial Intelligence
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will provide some help in setting this up at the start of the course.
Lectures may be recorded; these recordings will only be used within the course, including possibly in subsequent years as background information.

Recommended background knowledge

Some experience with the linux command line is preferred.

Symplectic Geometry

Course Code	XMM_418036
Credits	8
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

See https://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See https://elo.mastermath.nl

Method of Assessment

See https://elo.mastermath.nl

Additional Information Target Audience

mMath

Additional Information

The course is taught once every two years, the next opportunity will be in study year 2018-2019

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Systems and Control

Course Code	XMM_400332
Credits	6
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

See https://elo.mastermath.nl

Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

See https://elo.mastermath.nl

Method of Assessment

See https://elo.mastermath.nl

Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Topological Data Analysis

Course Code	XMM_0036
Credits	8
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course viahttps://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Tracer Kinetic Modelling

Course Code	XM_0014
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. M.M. Yaqub
Examiner	dr. ir. M.M. Yaqub
Teaching Staff	dr. S.V.S. Golla, dr. ir. M.M. Yaqub
Teaching method(s)	Seminar, Practical, Lecture

Course Objective

The 6 EC master course Advanced Medical Technology aims to provide the student knowledge and skills in the field of Tracer Kinetic Modelling. Starting with the basis of PET-imaging and signal-&-system-analysis the courses

develops to an state-of-the-art discussion of

1. different tracer-kinetic models and their physiological interpretation,
2. signal-&-system-analysis based administration (e.g. bolus infusion & steady state) of radiotracers and radiotracer concentration measurements,
3. estimation of model parameters using different least square estimation techniques.

Course Content

This year the course Advanced Medical Technology concerns the topic of Tracer Kinetic Modelling applied to Analysis of PET studies. To be short, a PET image visualises the spatial distribution of a radio-pharmaceutical (or radiotracer i.e., radio-labelled substances) within the human body. The quantitative analysis of the radiotracer distribution and its temporal behaviour, using the TracerKinetic-Modelling approach, enables the interpretation of the PET studies in terms of physiological processes, such as regional perfusion of the myocardium and brain, glucose utilisation in tissues and tumors, and binding of medicines at receptors in the cell membrane. Topics highly value for clinical diagnostics and, moreover, topics in state-of-the-art fundamental and clinical research. Tracer Kinetic Modelling is the field of studying the time course of a radiotracer and its metabolites in the human body by using quantitative methods such as: the Positron Emission Tomograph (PET), mathematical modelling of pharmacokinetics (using differential equations and the theory of signals & system), estimation of model parameter (using linear and non-linear least-square techniques) and, finally, the tools developed for the tracer-kinetic-model-based analysis and the interpretation and the clinical use of PET-images.

Additional Information Teaching Methods

The course consists of lectures and hands-on computer practicals with assignments and a final exam.

Method of Assessment

Hands-on practicals with assignments & written examination. No scoring is given for the hands-on practicals with assignments but students need to attend all hands-on sessions in order to be allowed to attend the exam. The final score for the course is based on just the exam score.

Explanation Canvas

All important information is communicated through canvas announcements.

Recommended background knowledge

Good to have read about medical imaging techniques before hand (bachelor level). However all needed theory will be explained during the course.

Tweetalig Onderwijs

Course Code	O_MTTO
Credits	3
Period	P2
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S. Donszelmann
Examiner	dr. S. Donszelmann
Teaching Staff	dr. S. Donszelmann
Teaching method(s)	Seminar

Course Objective

Na afloop van het vak heb je een beeld van het Tweetalig Onderwijs (TTO) binnen het Nederlandse voortgezet onderwijs en beschik je over basale vaardigheden om de Engelse taal op een zinvolle manier als instructietaal én als leertaal in te zetten.

Course Content

Op hoe langer hoe meer scholen wordt tweetalig onderwijs aangeboden (TTO). Het doel van dit type onderwijs is dat leerlingen tegelijk met vakinhoud ook Engels leren. Soms pakt dit goed uit, maar het kan ook zijn doel voorbij

schieten. Hoe zit dit?

Het gebruik van de doeltaal (Engels in dit geval) leidt niet bij voorbaat tot het leren van Engels. Leerzaam inzetten van de doeltaal is gebonden aan allerlei didactische voorwaarden waarvan de docent zich vaak maar beperkt bewust is. Het gevolg kan heel goed zijn dat leerlingen bij veel vakken nauwelijks Engels bijleren ten opzichte van het Engels dat zij al beheersten, en bovendien minder vakinhoud meekrijgen dan in het Nederlands het geval zou zijn omdat de docent zich omslachtig in een vreemde taal uitdrukt. Doeltaal-Leertaal onderzoek (Dönszelmann, 2019) geeft inzicht in de vraag welk didactisch instrumentarium er nodig is om leerlingen op leerzame manier onder te dompelen in de vreemde taal, terwijl er tegelijkertijd ook vakinhoud wordt overgebracht.

Binnen het keuzevak onderzoeken we wat dit betekent voor de TTO-praktijk binnen Nederlandse VO-scholen, en hoe goed TTO-onderwijs eruit moet zien.

Additional Information Teaching Methods

Werkcolleges

Method of Assessment

Kort literatuurverslag, of onderzoek binnen de eigen praktijk (als hier TTO-onderwijs voorkomt).

Entry Requirements

Didactiek 2, Praktijk 1

Literature

Wordt via Canvas bekend gemaakt.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten

Custom Course Registration

Er geldt een maximum van 5 voor bijvakstudenten van andere lerarenopleidingen.

Uncertainty Quantification

Course Code	XM_0183
Credits	6
Period	P1+2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.B. Dubinkina
Examiner	dr. S.B. Dubinkina
Teaching Staff	dr. S.B. Dubinkina
Teaching method(s)	Seminar

Course Objective

At the end of this course, the student is able

- to learn part of the mathematical material on their own, and to give a talk about it;
- to conduct a discussion about material presented by someone else;
- to implement a research idea into a MATLAB code;
- to analyse the obtained results;
- to write a report;

Course Content

The aim of this master seminar is to learn how to read scientific texts at a research level, create and organise

material, to give seminar talks to your fellow students, and to be able to explain the mathematical concepts and ideas involved. It is also a goal to learn how to constructively participate in a research seminar by asking questions and giving feedback. Finally, a student will implement a research idea into a MATLAB code and write a comprehensive report

In this master seminar, we focus on uncertainty quantification and data assimilation. This is an ongoing research field that combines dynamical models and data. Applications are weather forecasting, water subsurface estimations, epidemiology, to name a few.

Additional Information Teaching Methods

A seminar, for 2 hours per week during each week of periods 1 and 2. Each student gives one or two talks. Furthermore, at the end of the course the student transfers a mathematical method into a MATLAB code and writes a comprehensive report.

Method of Assessment

Presentations, MATLAB code, and a report will contribute to the total grade. The weights will be specified on Canvas. There will be a minimum attendance requirement in order to get a grade for this course. Due to the nature of the seminar, there will not be a resit.

Literature

Probabilistic Forecasting and Bayesian Data Assimilation by Sebastian Reich and Colin Cotter, ISBN 1107663911

Additional Information

If the number of participating students is too low to have a presentation every week, we might change some sessions into discussions of material that has to be read in advance.

Vernieuwende onderwijsconcepten

Course Code	O_MLVNON
Credits	3
Period	P2, P5
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. M. Meeter
Examiner	prof. dr. M. Meeter
Teaching Staff	prof. dr. M. Meeter, dr. H.B. Westbroek
Teaching method(s)	Seminar, Lecture

Course Objective

Na het volgen van dit keuzevak, kun je:

1. beschrijven hoe het onderwijs in Nederland de afgelopen eeuwen vorm gekregen heeft en hoe met het onderwijs antwoord gegeven is op de vragen: wat willen we onze kinderen leren; waartoe dient onderwijs en hoe geven we onderwijs vorm?
2. een aantal vernieuwende vormen van VO-onderwijs opsommen (traditionele vernieuwende onderwijsconcepten, zoals Montessori, Dalton, Vrije School en hedendaagse vernieuwende onderwijsconcepten zoals gepersonaliseerd leren/Kunskapsskolan, Agora) en beschrijven wat de ideeën achter de vernieuwingen zijn.
3. een eigen leervraag over vernieuwende onderwijsconcepten en/of processen die daarbij horen formuleren en vragen bedenken die je tijdens het schoolbezoek aan docenten en leerlingen wilt stellen om de leervraag te kunnen beantwoorden.
4. de eigen leervraag beantwoorden met gebruikmaking van theorie, gesprekken met docenten en leerlingen, de bijeenkomsten en het schoolbezoek en een onderbouwd antwoord geven op de vraag: Wat vind ik belangrijk bij het onderwijs dat ik verzorg? Wat spreekt mij aan?
5. een begrippenkader hanteren waarmee je kunt nadenken en praten over vernieuwingsprocessen en vernieuwende onderwijsconcepten.
6. een beredeneerd oordeel vormen en een professionele dialoog voeren over voor- en nadelen van vernieuwingen en ongeteste aannames en kritische implementatiedetails benoemen.

Course Content

In het Nederlandse onderwijs, zeker in competitieve markten als de Amsterdamse, wordt continu vernieuwd. Scholen sturen bijvoorbeeld aan op het implementeren van gepersonaliseerd leren of projectmatig werken of onderwijs in gemengde stamgroepen. Bij dit vak bezoeken we een aantal vernieuwingsscholen en analyseren we de beweegredenen en de proceskant van onderwijsvernieuwing (de implementatie).

Additional Information Teaching Methods

Werkcolleges (waarin de dialoog centraal staat en waarin je collega's uit het de scholen interviewt) en schoolbezoeken.

Method of Assessment

Analyse van een schoolvernieuwing in de vorm van een academisch essay of in de vorm van een podcast of educlip (keuze) aan de hand van wetenschappelijke- en vakliteratuur, interviews, observaties en schoolbezoeken.

Entry Requirements

Didactiek 1 en 2, Praktijk 1

Literature

Informatie over te lezen artikelen wordt gedeeld op Canvas.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Wereldburgerschap deel 1

Course Code	O_MWRLDBS_1
Credits	3
Period	P1, P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	D.S. Hoefakker-Laanstra, dr. I. Pauw, drs. H.R. Goudsmit
Teaching method(s)	Study Group

Course Objective

Na het volgen van dit keuzevak:

- kan je het belang van wereldburgerschap voor leerlingen, docenten en scholen onderbouwd over het voetlicht brengen.
- ken je de verschillende definities van wereldburgerschap en kan je op grond daarvan een onderbouwde eigen formulering geven.
- kan je de verschillende vormen van wereldburgerschap noemen en kan je bij de verschillende vormen leerlijnen en concrete activiteiten noemen.
- kan je aan de hand van de gekozen definitie en de methodologie van wereldburgerschap een onderbouwde leerlijn wereldburgerschap voor een school ontwerpen
- kan je in de klas in het kader van wereldburgerschap een maatschappelijke controverse les ontwerpen en uitvoeren .
- kan je in je klas met leerlingen met behulp van de dialoog maatschappelijke thema’s bespreken

Course Content

Leerlingen van nu groeien op in een gepolariseerde multiculturele omgeving in een globaliserende wereld, waarin belangrijke thema’s als klimaat, de energietransitie en ook oorlog en vrede de landsgrenzen overstijgen. De

samenleving van nu vraagt nieuwe en andere competenties van leerlingen. Zo is het belangrijk, dat leerlingen in dialoog met elkaar kunnen gaan in plaats van in debat. En moeten leerlingen zich een aanpak eigen maken om de grote maatschappelijke thema's voor zichzelf behapbaar te maken. Ook vraagt het van leerlingen om een open en onderzoekende houding ten opzichte van andere meningen en ideeën. Anders gezegd, de maatschappij vraagt om leerlingen, die zich zowel nu in hun eigen omgeving als later in de maatschappij kunnen ontwikkelen tot wereldburgers . En het is de taak van het onderwijs hen handvatten mee te geven om wereldburgers te worden. Hoe je als docent en als school dat kan doen is het onderwerp van deze specialisatie.

Additional Information Teaching Methods

Werkcolleges

Method of Assessment

Verschillende toetsvormen: leerlijn Wereldburgerschap ontwerpen, controverse les ontwerpen en uitvoeren

Entry Requirements

Didactiek 1, Praktijk 1

Literature

- Wereldgericht Onderwijzen.Biesta in praktijk, B.Eidhof e.a, 2021
- Artikelen via Canvas

Wereldburgerschap Deel 2

Course Code	O_MWRLDBS_2
Credits	3
Period	P2, P5
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	dr. I. Pauw, drs. H.R. Goudsmit, D.S. Hoefakker-Laanstra
Teaching method(s)	Seminar, Lecture

Course Objective

Doelen.

- Studenten kennen de verschillende internationale onderzoeken naar de diverse schoolsystemen, kunnen de indicatoren van de verschillende onderzoeken aangeven en zijn in staat tot het geven van een onderbouwde kijk op deze onderzoeken.
- Studenten kennen de kritische momenten in een schoolsysteem en zijn in staat deze momenten te waarderen
- Studenten zijn in staat andere onderwijssystemen dan het Nederlandse in kaart te brengen en deze te relateren aan het eigen onderwijsstelsel
- Studenten kunnen het begrip Internationalisering definieren, de mate van Internationalisering op hun eigen school in kaart brengen en een leerlijn Internationalisering voor een school ontwerpen
- Studenten kunnen hun visie op goed onderwijs verdiepen naar aanleiding van analyseren van andere schoolsystemen. Daarnaast zijn ze ook in staat een meer onderbouwde kijk op het Nederlands onderwijssysteem te geven

Course Content

Onlangs sprak een bekende Nederlandse publicist naar aanleiding van hevige discussies rond de klimaatcrisis de gevleugelde woorden: "Velen hebben een mening, slechts weinigen komt een oordeel toe." Nederlanders staan er om bekend overal een mening over te hebben en die te pas en te onpas te ventileren.

Je kan je afvragen of dat ook niet voor het onderwijs geldt. Er wordt immers veel gezegd en geschreven over ons onderwijs zonder dat het altijd duidelijk waarop het geventileerde is gebaseerd.

En hoewel wij er dagelijks deel van uit maken is het maar de vraag hoe onderbouwd onze eigen visie is op het Nederlands onderwijssysteem. Wordt onze eigen kijk ook niet teveel ingegeven door een beperkte hoeveelheid

aan eigen ervaringen en missen wij niet het overzicht of de helikopter view om te komen tot een gefundeerd oordeel? Missen wij ook niet voldoende vergelijkingsmateriaal om tot een werkelijk gefundeerd oordeel te komen? Deze keuzemodule wil dan ook in de eerste plaats een stimulans zijn om tot een dergelijk verdiepte visie te komen.

Overal in de wereld tonen regeringen en politici interesse in de uitkomsten van de benchmarks, die internationaal worden georganiseerd door instellingen als de OESO (PISA; Eduaction at Glance). De uitkomsten van deze rapporten zorgen in de deelnemende landen vaak voor onderwijshervormingen. Maar het doen van internationaal vergelijkend onderzoek roept ook veel discussie op. Kan je onderwijssystemen wel vergelijken op grond van cijfers, zijn de onderzoeken wel betrouwbaar en valide, kun je onderwijssystemen uberhaupt wel met elkaar vergelijken? In dit keuzevak zullen we dan ook ingaan op de verschillende onderzoeken, onderzoeksmethodes en kritieken. Maar ook gaan we zelf (een klein) onderzoek doen naar de verschillende onderwijssystemen om te kijken of we ons een beeld kunnen vormen van andere schoolsystemen.

Dit keuzevak is enerzijds een aanvulling op deel 1 van Wereldburgerschap maar is ook als "los"keuzevak te volgen

Additional Information Teaching Methods

Werkcollege

Method of Assessment

Het schrijven van een essay

Entry Requirements

Didactiek 1 en praktijk 1

Literature

Artikelen uitgereikt op college

Recommended background knowledge

Didactiek 1 en Praktijk 1